UNIT-III

Network Layer

Network Layer

Network layer is majorly focused on getting packets from the source to the destination, routing error handling and congestion control.

It’s various functions include:

- **Addressing:**
  Maintains the address at the frame header of both source and destination and performs addressing to detect various devices in network.

- **Packeting:**
  This is performed by Internet Protocol. The network layer converts the packets from its upper layer.

- **Routing:**
  It is the most important functionality. The network layer chooses the most relevant and best path for the data transmission from source to destination.

- **Inter-networking:**
  It works to deliver a logical connection across multiple devices.

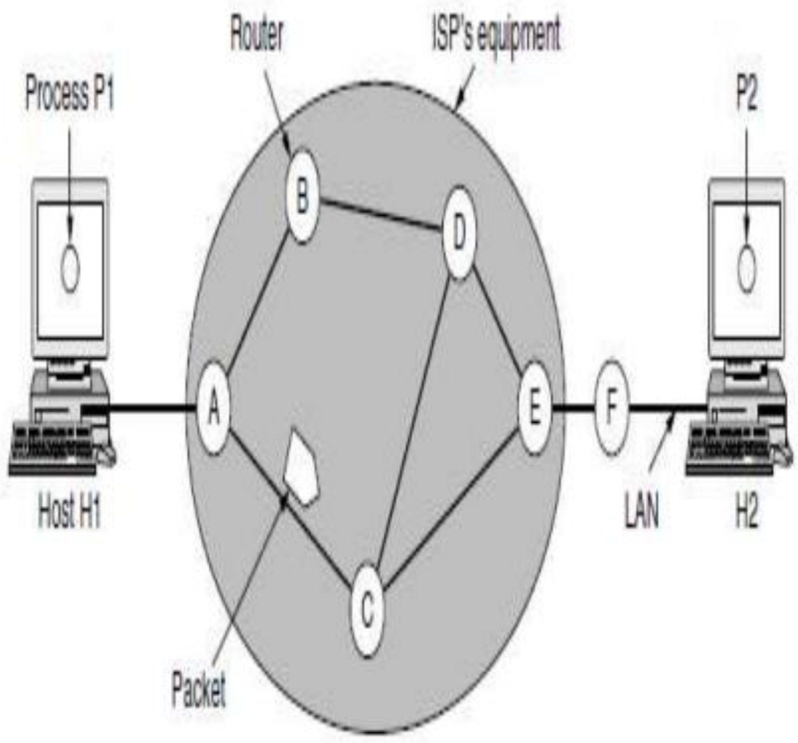
NETWORK LAYER DESIGN ISSUES

The network layer comes with some design issues they are described as follows:

- Store-and-Forward Packet Switching
- Services Provided to the Transport Layer
- Implementation of Connectionless Service
- Implementation of Connection-Oriented Service
- Comparison of Virtual-Circuit and Datagram Networks

1. Store and Forward packet switching

The host sends the packet to the nearest router. This packet is stored there until it has fully arrived once the link is fully processed by verifying the checksum then it is forwarded to the next router till it reaches the destination. This mechanism is called "Store and Forward packet switching."



2. Services Provided to the Transport Layer

Through the network/transport layer interface, the network layer transfers it's services to the transport layer.

Before providing these services to the transfer layer following goals must be kept in mind :-

- Offering services must not depend on router technology.
- The transport layer needs to be protected from the type, number and topology of the available router.
- The network addresses for the transport layer should use uniform numbering pattern also at LAN and WAN connections.

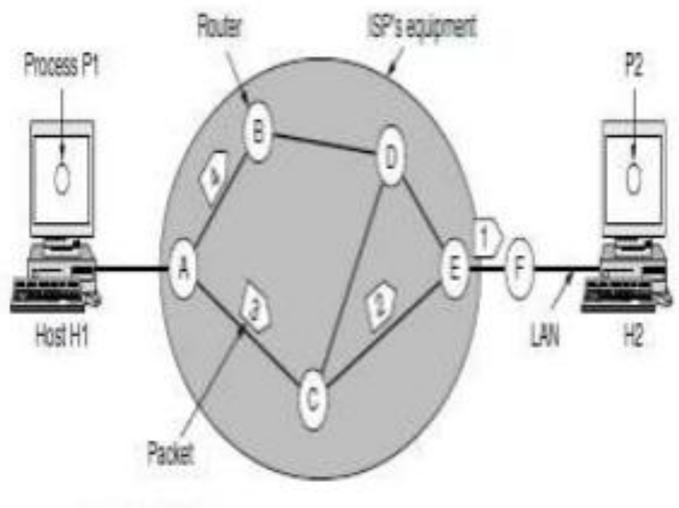
Based on the connections there are 2 types of services provided:

Connectionless – Connectionless service is a data transmission method used in packet switching networks by which each data unit is individually addressed and routed based on information carried in each unit, rather than in the setup information of a prearranged, fixed data channel as in connection-oriented communication.

Connection-Oriented – Connection-oriented service is a network communication mode, where a communication session or a semi-permanent connection is established before any useful data can be transferred, and where a stream of data is delivered in the same order as it was sent.

3. Implementation of Connectionless Service

Packet are termed as “datagrams” and corresponding subnet as “datagram subnets”. When the message size that has to be transmitted is 4 times the size of the packet, then the network layer divides into 4 packets and transmits each packet to router via. a few protocol. Each data packet has destination address and is routed independently irrespective of the packets.



4. Implementation of Connection Oriented service

To use a connection-oriented service, first we establishes a connection, use it and then release it. In connection-oriented services, the data packets are delivered to the receiver in the same order in which they have been sent by the sender.

It can be done in either two ways:

- Circuit Switched Connection – A dedicated physical path or a circuit is established between the communicating nodes and then data stream is transferred.
- Virtual Circuit Switched Connection – The data stream is transferred over a packet switched network, in such a way that it seems to the user that there is a dedicated path from the sender to the receiver. A virtual path is established here. While other connections may also be using the same path.

Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Routing Vs Forwarding

- **Forwarding** refers to a device sending a datagram to the next device in the path to the destination.
- **Routing** refers to the process a layer-3 device uses to decide on what to do with a layer-3 packet.
- The whole process a router uses is called **routing**, but within the router, it **switches** packets before **forwarding** them to the next hop device.

Static Routing

A routing table can either be static and dynamic.

- Static Routing is also known as **non-adaptive** routing which doesn’t change routing table unless the network administrator changes or modify them manually. Static routing does not use complex routing algorithms and It provides high or more security than dynamic routing.

Dynamic Routing

- Dynamic routing is also known as **adaptive** routing which change routing table according to the change in topology. Dynamic routing uses complex routing algorithms, and it does not provide high security like static routing. When the network change(topology) occurs, it sends the message to router to ensure the changes and then the routes are recalculated for sending updated routing information.

Difference between Static and Dynamic Routing:

STATIC ROUTING	DYNAMIC ROUTING
In static routing routes are user defined.	In dynamic routing, routes are updated according to topology.
Static routing does not use complex routing algorithms.	Dynamic routing uses complex routing algorithms.
Static routing provides high or more security.	Dynamic routing provides less security.
Static routing is manual.	Dynamic routing is automated.
Static routing is implemented in small networks.	Dynamic routing is implemented in large networks.
In static routing, additional resources are not required.	In dynamic routing, additional resources are required.
In static routing, failure of link disrupts the rerouting.	In dynamic routing, failure of link does not interrupt the rerouting.

Types of Routing

- If a datagram is destined for only one destination (one-to-one delivery), we have **unicast routing.**
- If the datagram is destined for several destinations (one-to-many delivery), we have **multicast routing.**
- Routing can be possible if a router has a forwarding table to forward a packet to the appropriate next node on its way to the final destination or destinations.

Uni-cast Routing

Unicast means the transmission from a single sender to a single receiver. It is a point-to-point communication between sender and receiver.

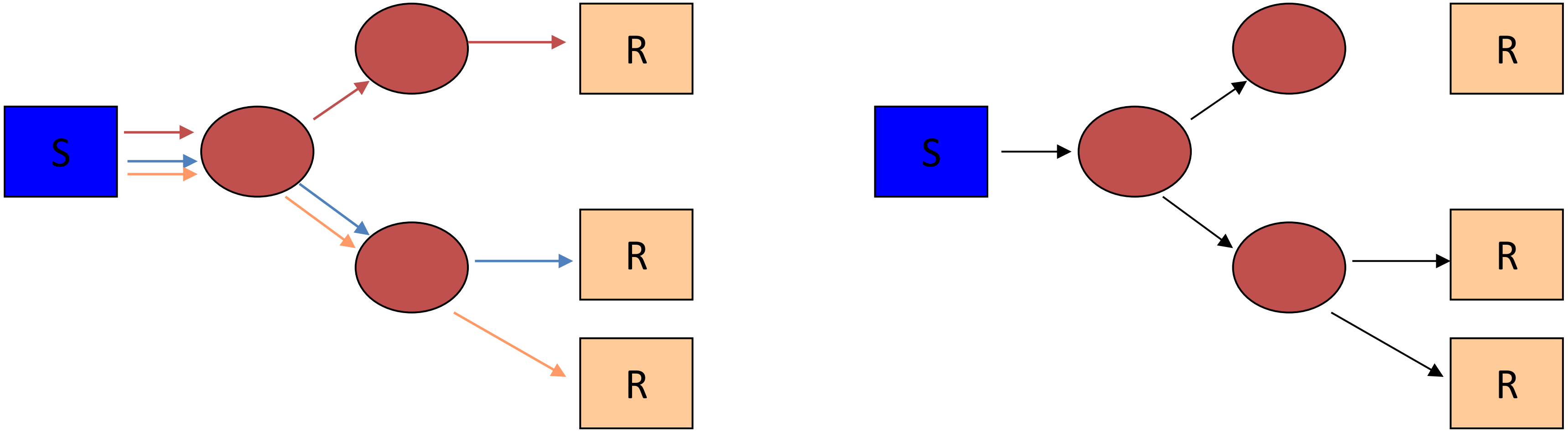
TCP is the most commonly used **unicast protocol**.

It is a connection-oriented **protocol** that relay on acknowledgement from the receiver side.

Multicast Routing

A **Multicast** transmission sends IP packets to a group of hosts on a network.

- Application level one to many communication
- multiple unicasts



Why Multicast

- When sending same data to multiple receivers
  - better bandwidth utilization
  - less host/router processing
  - quicker participation
- Application
  - Video/Audio broadcast (One sender)
  - Video conferencing (Many senders)
  - Real time news distribution
  - Interactive gaming

Routing Algorithms

- Properties
- Shortest Path Routing
- Flooding
- Distance Vector Routing
- Link State routing
- Hierarchical routing
- Broadcast routing
- Multicast routing
- Routing for mobile hosts
- Routing in Ad Hoc Networks
- Node Lookup in Peer-to-Peer Networks

Most important algorithms!

Properties Of Routing Algorithm

Routing is the process of forwarding of a packet in a network so that it reaches its intended destination.

•Correctness**:** The routing should be done properly and correctly so that the packets may reach their proper destination.

•Simplicity**:** The routing should be done in a simple manner so that the overhead is as low as possible. With increasing complexity of the routing algorithms the overhead also increases.

•Robustness**:** Once a major network becomes operative, it may be expected to run continuously for years without any failures. The algorithms designed for routing should be robust enough to handle hardware and software failures and should be able to cope with changes in the topology and traffic without requiring all jobs in all hosts to be aborted and the network rebooted every time some router goes down.

- Stability: The routing algorithms should be stable under all possible circumstances.

- Fairness: Every node connected to the network should get a fair chance of transmitting their packets. this is generally done on a first come first serve basis.

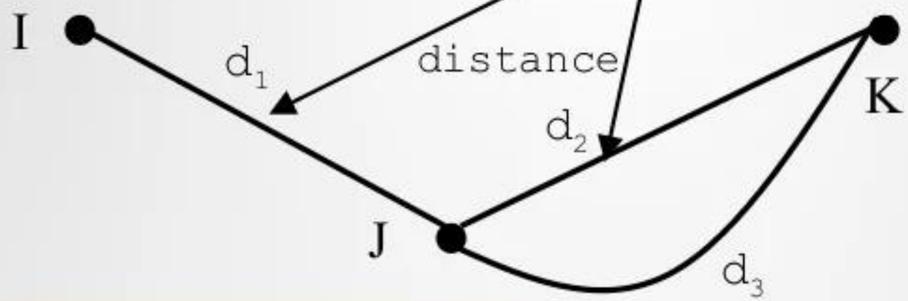
- Optimality: The routing algorithms should be optimal in terms of throughput and minimizing mean packet delays. here there is a trade-off and one has to choose depending on his suitability.

Types Of Routing Algorithms

- **Nonadaptive (Static)**
  - Do not use measurements of current conditions
  - Static routes are downloaded at boot time
- **Adaptive Algorithms**
  - Change routes dynamically
    - Gather information at runtime
      - locally
      - from adjacent routers
      - from all other routers
    - Change routes
      - Every delta T seconds
      - When load changes
      - When topology changes

Optimality Principle

Optimal path from I to K over J



$d_1 + d_2$ is minimal

Other path from J to K

$d_3 > d_2$

as

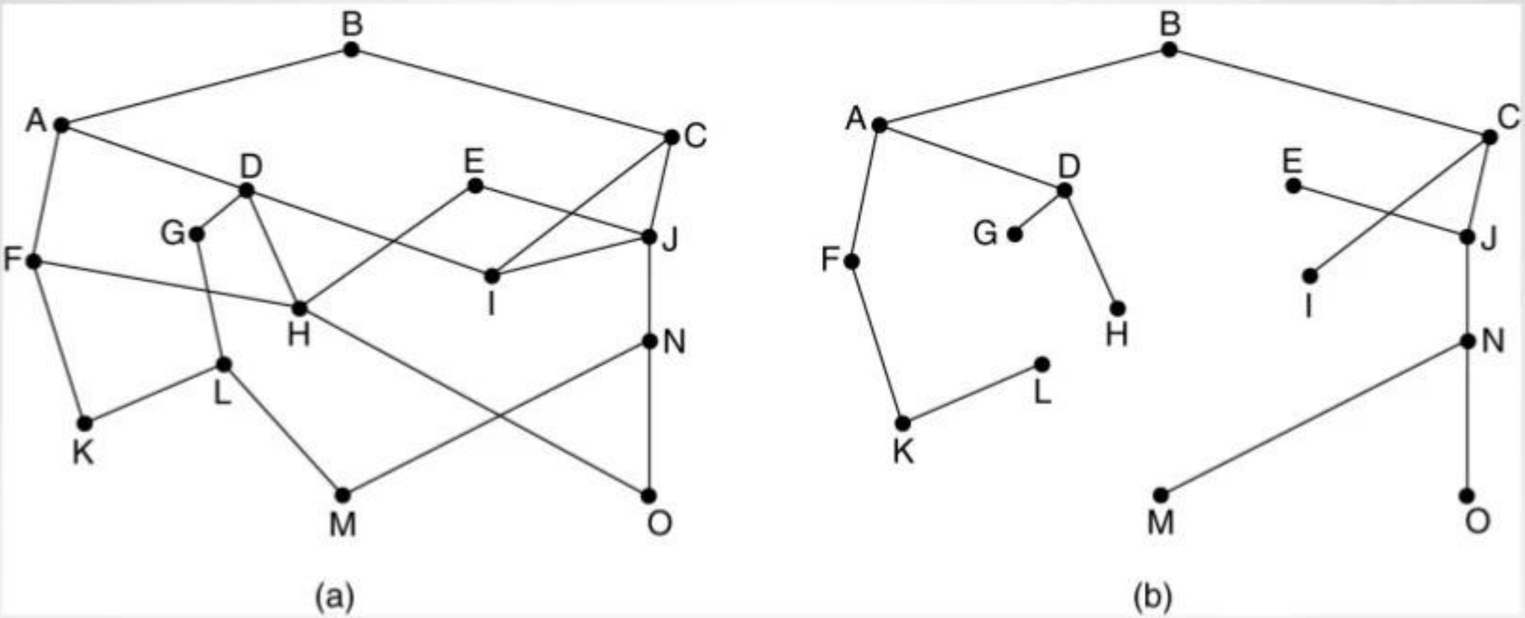
$d_1 + d_3 > d_1 + d_2$

Set of all optimal routes

- from all sources
- to a given destination

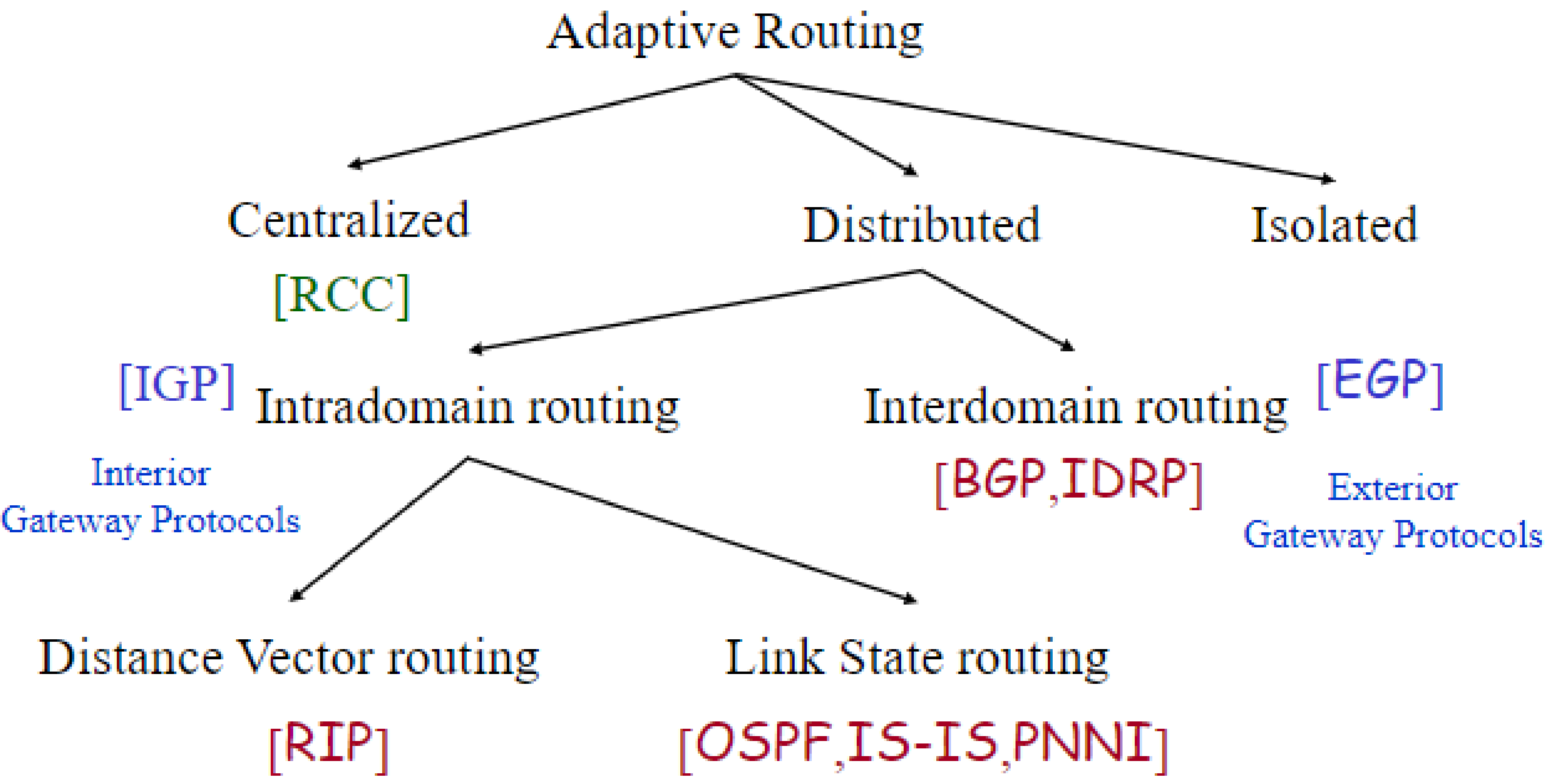
is a tree: sink tree

- The set of optimal routes to a particular node forms a sink tree.
- Sink trees are not necessarily unique.
- Goal of all routing algorithms
  - Discover sink trees for all destinations



(A) A SUBNET. (B) A SINK TREE FOR ROUTER B.

Internetwork Routing [Halsall]



Least-Cost Routing

- The best route from the source router to the destination router is to find the least cost between the two

- In other words, the source router chooses a route to the destination router in such a way that the total cost for the route is the least cost among all possible routes

Least-Cost Trees

- If there are N routers in an internet, there are (N – 1) least-cost paths from each router to any other router.
- This means we need N × (N – 1) least-cost paths for the whole internet.
- If we have only 10 routers in an internet, we need ?????? least-cost paths.
- **<u>A least-cost tree</u>** is a tree with the source router as the root that spans the whole graph (visits all other nodes) and in which the path between the root and any other node is the shortest.

Least cost tree for the given below graph will be as shown below:

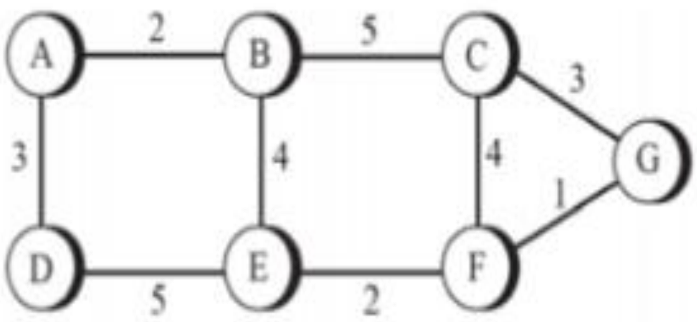
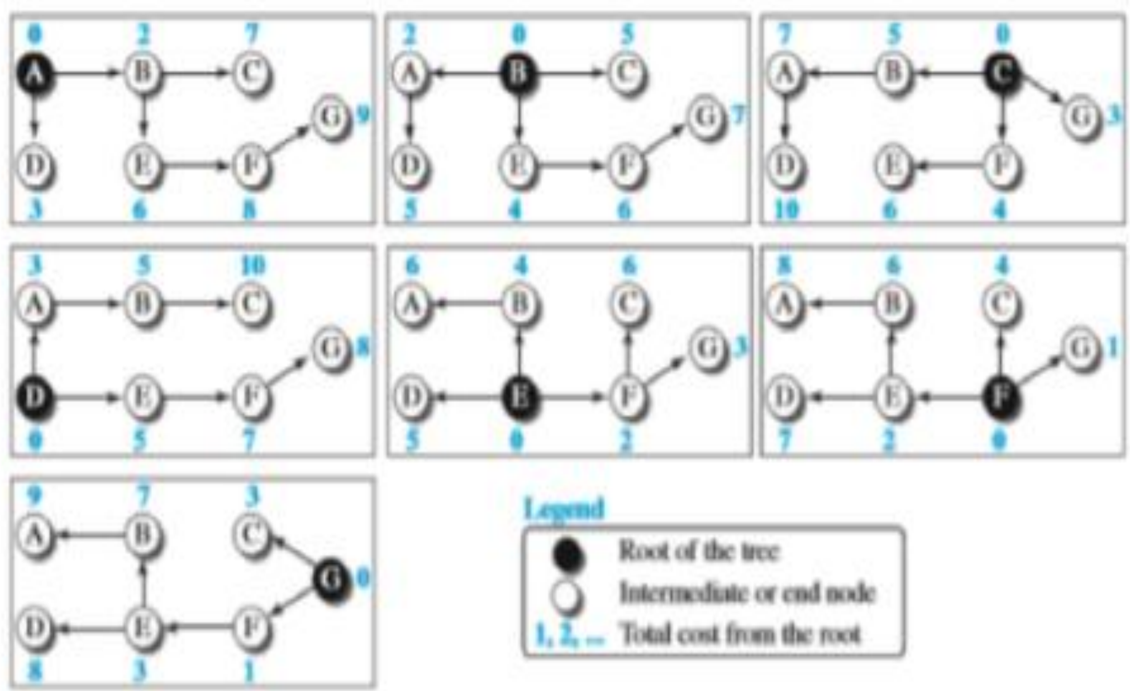


Figure 20.2 *Least-cost trees for nodes in the internet of Figure 20.1*



Shortest Path Routing

1. Bellman-Ford Algorithm [Distance Vector]
2. Dijkstra's Algorithm [Link State]

What does it mean to be the shortest (or optimal) route?

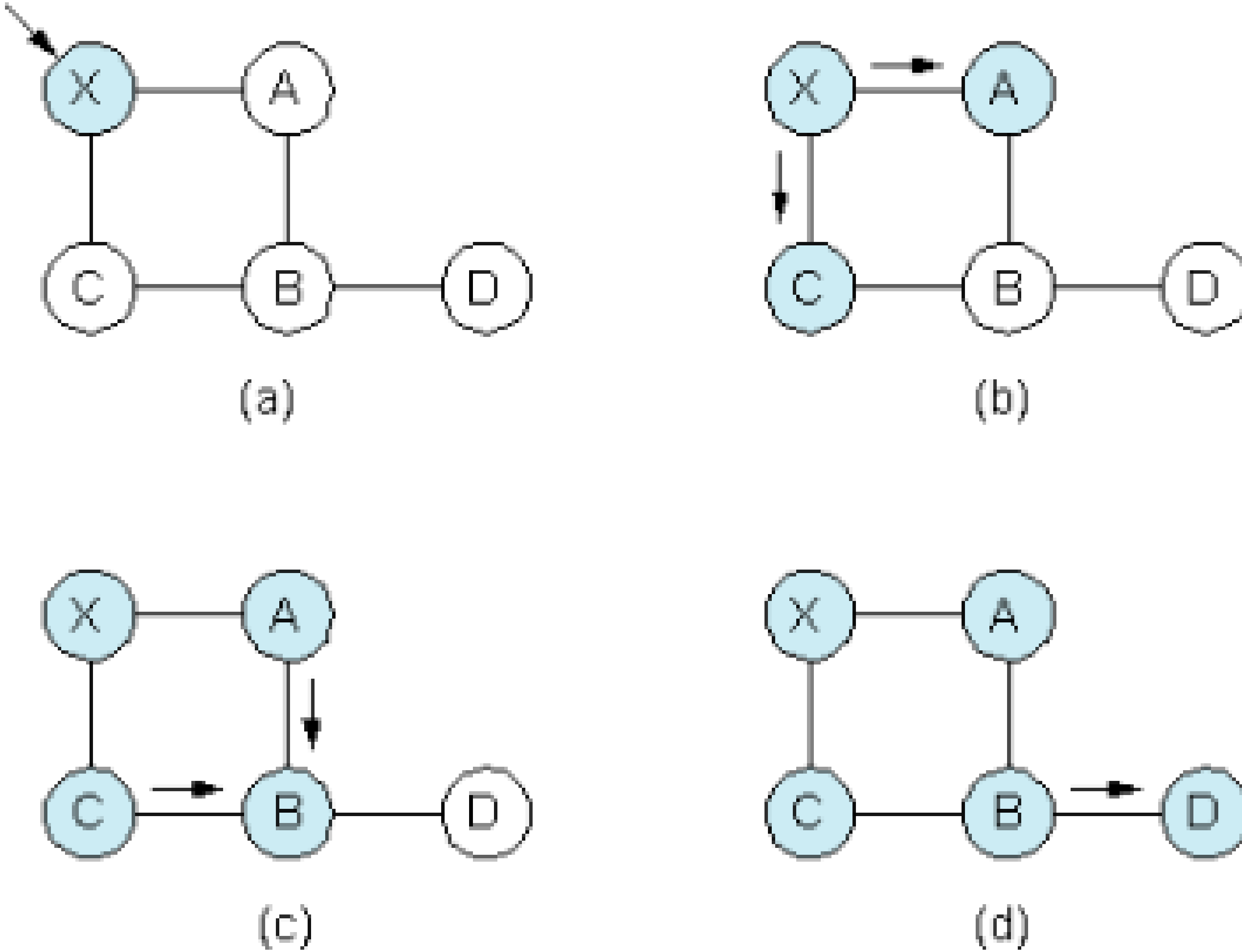
a. Minize the number of hops along the path.
b. Minimize mean packet delay.
c. Maximize the network throughput.

Link State Algorithm

1. Each router is responsible for meeting its neighbors and learning their names.
2. Each router constructs a link state packet **(LSP)** which consists of a list of names and cost to reach each of its neighbors.
3. The **LSP** is transmitted to ***ALL other routers***. Each router stores the most recently generated **LSP** from each other router.
4. Each router uses complete information on the network topology to compute the *shortest path route* to each destination node.

Link State Routing Protocol

- Link state routing protocols maintain complete road map of the network in each router running a link state routing protocol.
- Link State Protocols relay specific link characteristics and state information
- Unlike Distance Vector Routing Protocol, in this updates are sent as well
- Each router running a link state routing protocol originates information about the router, its directly connected links, and the state of those links.
- It takes into account the states of links of all the routers in a network.
- This information is sent to all the routers in the network as multicast messages. Link-state routing always try to maintain full networks topology by updating itself incrementally whenever a change happen in network.


X
A
C
B
D
(a)
X
A
C
B
D
(b)
X
A
C
B
D
(c)
X
A
C
B
D
(d)

Distance Vector Routing (DVR) Protocol

- The term distance vector refers to the fact that the protocol manipulates vectors (arrays) of distances to other nodes in the network. A **distance-vector routing (DVR)** protocol requires that a router inform its neighbors of topology changes periodically. Each router maintains a Distance Vector table containing the distance between itself and ALL possible destination nodes. Distances, based on a chosen metric, are computed using information from the neighbors’ distance vectors.

Information kept by DV router -

- Each router has an ID
- Associated with each link connected to a router, there is a link cost (static or dynamic).
- Intermediate hops count

Distance Vector Table Initialization –

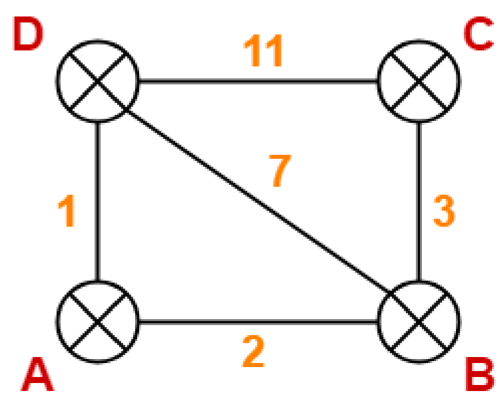
- Distance to itself = 0
- Distance to ALL other routers = infinity number.

Distance Vector Algorithm

- A router transmits its distance vector to each of its neighbors in a routing packet.
- Each router receives and saves the most recently received distance vector from each of its neighbors.
- A router recalculates its distance vector when:
  - It receives a distance vector from a neighbor containing different information than before.
  - It discovers that a link to a neighbor has gone down.

Destination	Distance	Next Hop
A	1	A
B	7	B
C	11	C
D	0	D

Destination	Distance	Next Hop
A	∞	–
B	3	B
C	0	C
D	11	D



Destination	Distance	Next Hop
A	0	A
B	2	B
C	∞	–
D	1	D

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	7	D

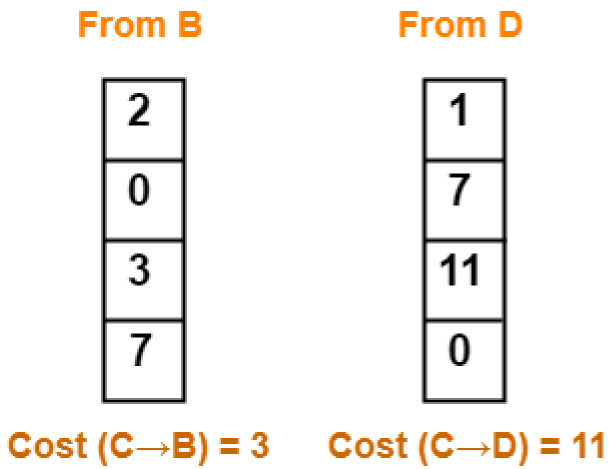
From B	From D
2	1
0	7
3	11
7	0

Cost(A→B) = 2   Cost(A→D) = 1

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

From A	From C	From D
0	∞	1
2	3	7
∞	0	11
1	11	0
Cost (B→A) = 2	Cost (B→C) = 3	Cost (B→D) = 7

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A



Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	10	B

From A	From B	From C
0	2	∞
2	0	3
∞	3	0
1	7	11
Cost (D→A) = 1	Cost (D→B) = 7	Cost (D→C) = 11

Destination	Distance	Next Hop
A	1	A
B	3	A
C	10	B
D	0	D

From B

$$\begin{bmatrix} 2 \\ 0 \\ 3 \\ 3 \end{bmatrix}$$

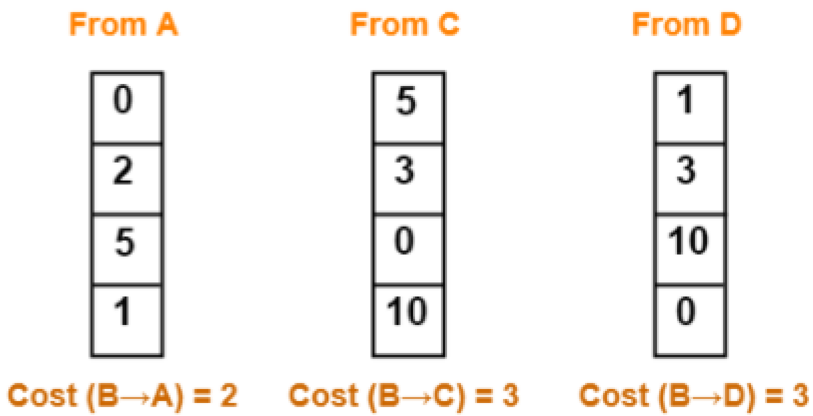
Cost(A→B) = 2

From D

$$\begin{bmatrix} 1 \\ 3 \\ 10 \\ 0 \end{bmatrix}$$

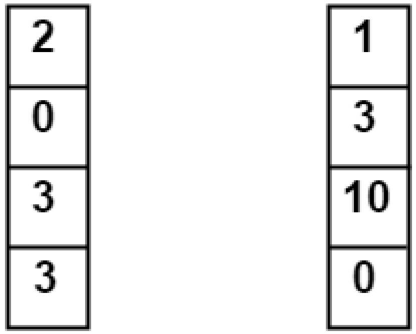
Cost(A→D) = 1

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D



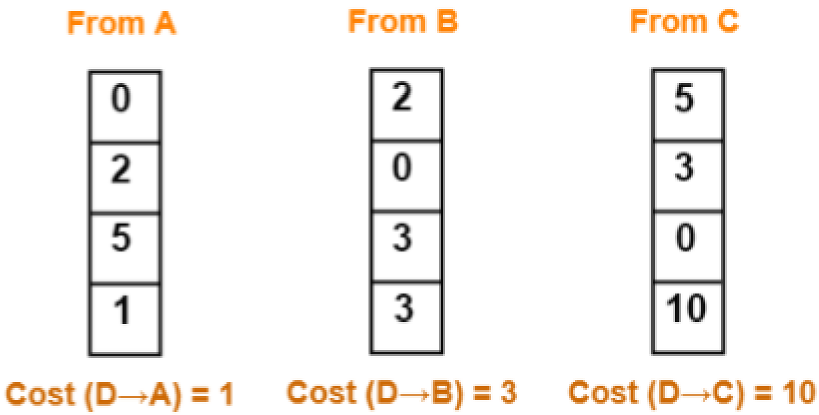
Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

From B From D



Cost (C→B) = 3 Cost (C→D) = 10

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	6	B



Destination	Distance	Next Hop
A	1	A
B	3	A
C	6	A
D	0	D

Count to infinity problem

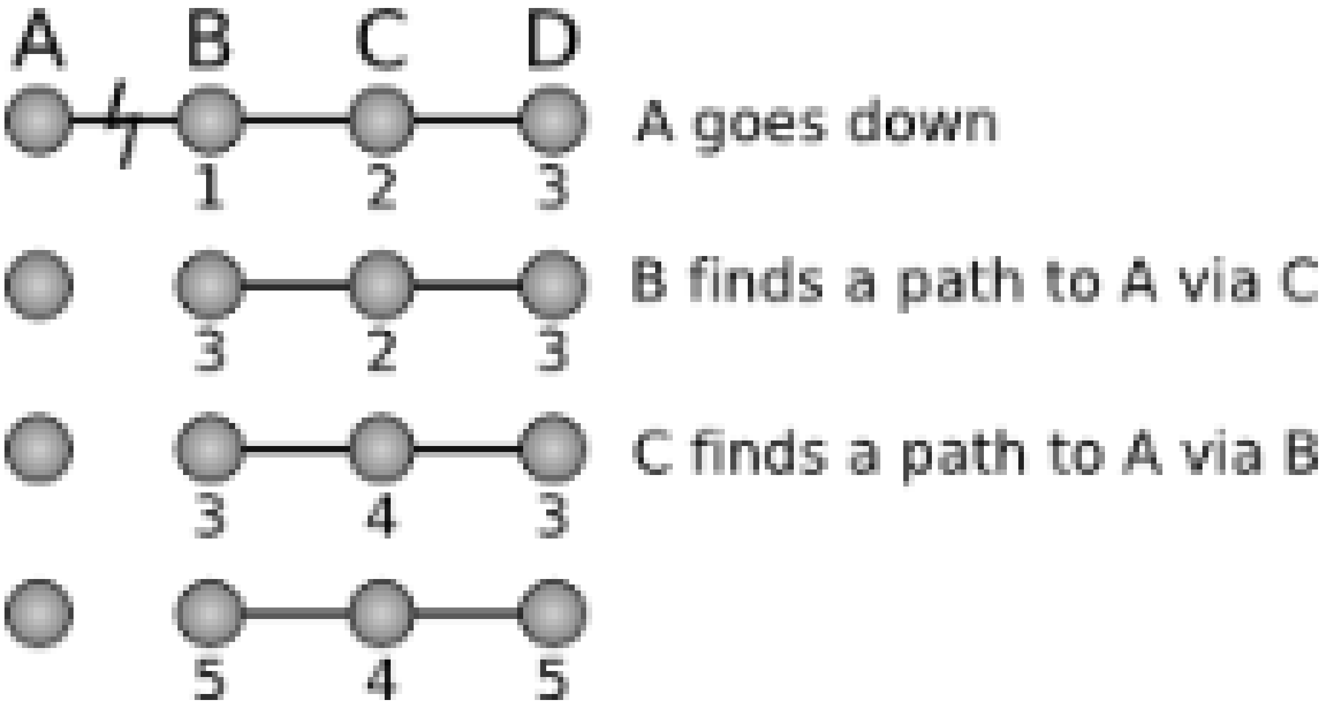


Figure 1: Count-to-infinity problem

Advantages of Distance Vector routing –

- It is simpler to configure and maintain than link state routing.

Disadvantages of Distance Vector routing –

- It is slower to converge than link state.
- It is at risk from the count-to-infinity problem.
- It creates more traffic than link state since a hop count change must be propagated to all routers and processed on each router. Hop count updates take place on a periodic basis, even if there are no changes in the network topology, so bandwidth-wasting broadcasts still occur.
- For larger networks, distance vector routing results in larger routing tables than link state since each router must know about all other routers. This can also lead to congestion on WAN links.

Shortest Path Routing

Shortest path routing refers to the process of finding paths through a network that have a minimum of distance or other cost metric.

Shortest Path Calculation: **Dijkstra's algorithm** can be used to determine the shortest path from one node in a graph to every other node within the same graph data structure, provided that the nodes are reachable from the starting node. Dijkstra's algorithm can be used to find the shortest path.

Dijkstra's Shortest Path Algorithm

Initially mark all nodes (except source) with infinite distance.

working node = source node

Sink node = destination node

While the working node is not equal to the sink

1. Mark the working node as permanent.
2. Examine all adjacent nodes in turn

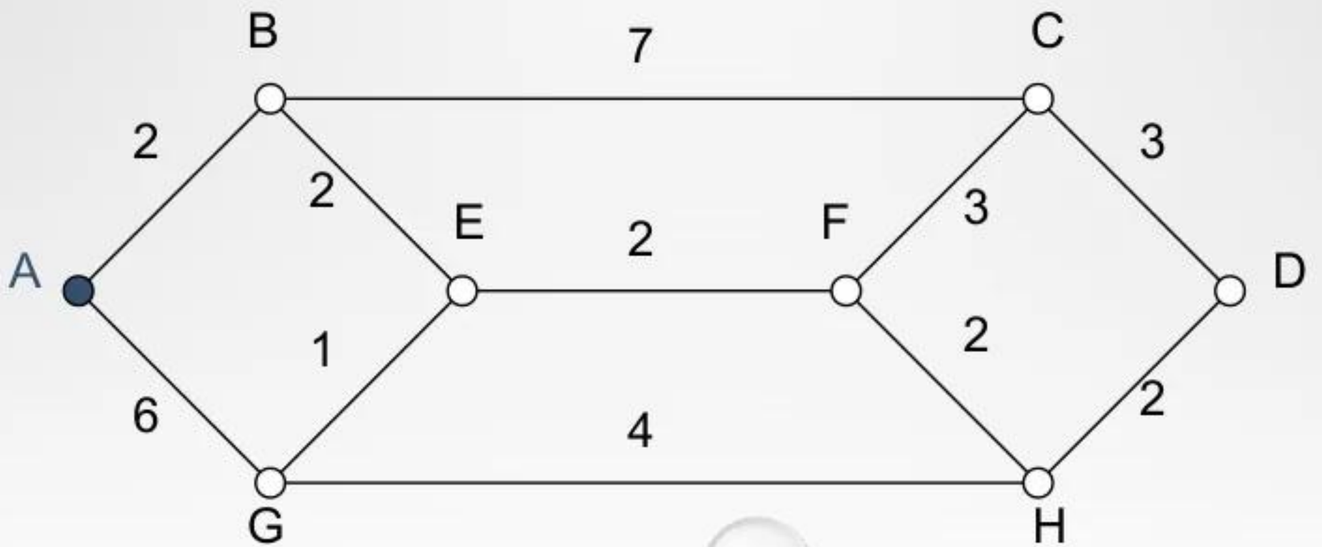
   If the sum of label on working node plus distance from working node to adjacent node is less than current labeled distance on the adjacent node, this implies a shorter path. Relabel the distance on the adjacent node and label it with the node from which the probe was made.

3. Examine all tentative nodes (not just adjacent nodes) and mark the node with the smallest labeled value as permanent. This node becomes the new working node.

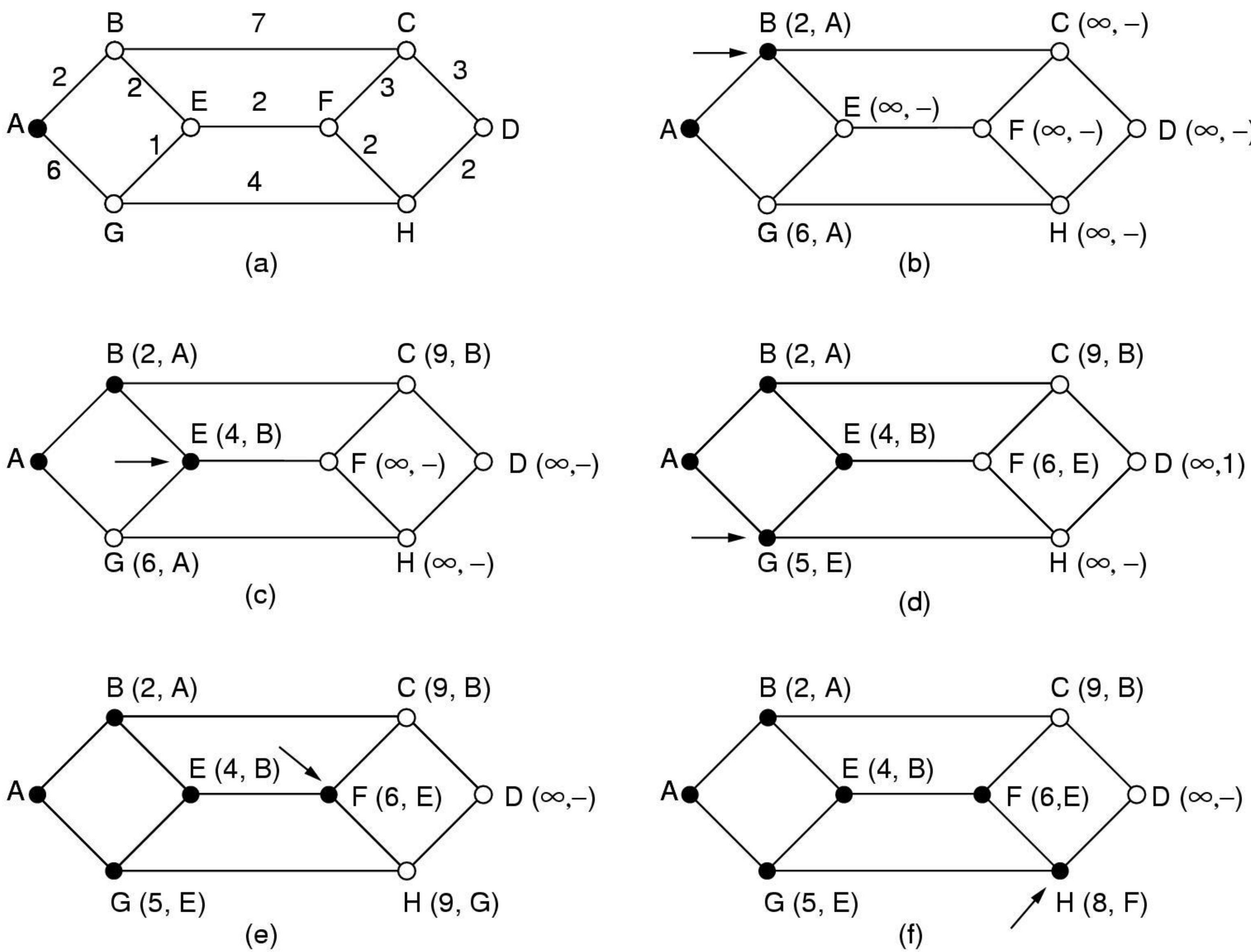
Reconstruct the path backwards from sink to source.

Dijkstra's Algorithm

Each node is labeled (in parentheses) with its distance from the source node along the best known path.



Routing: shortest path



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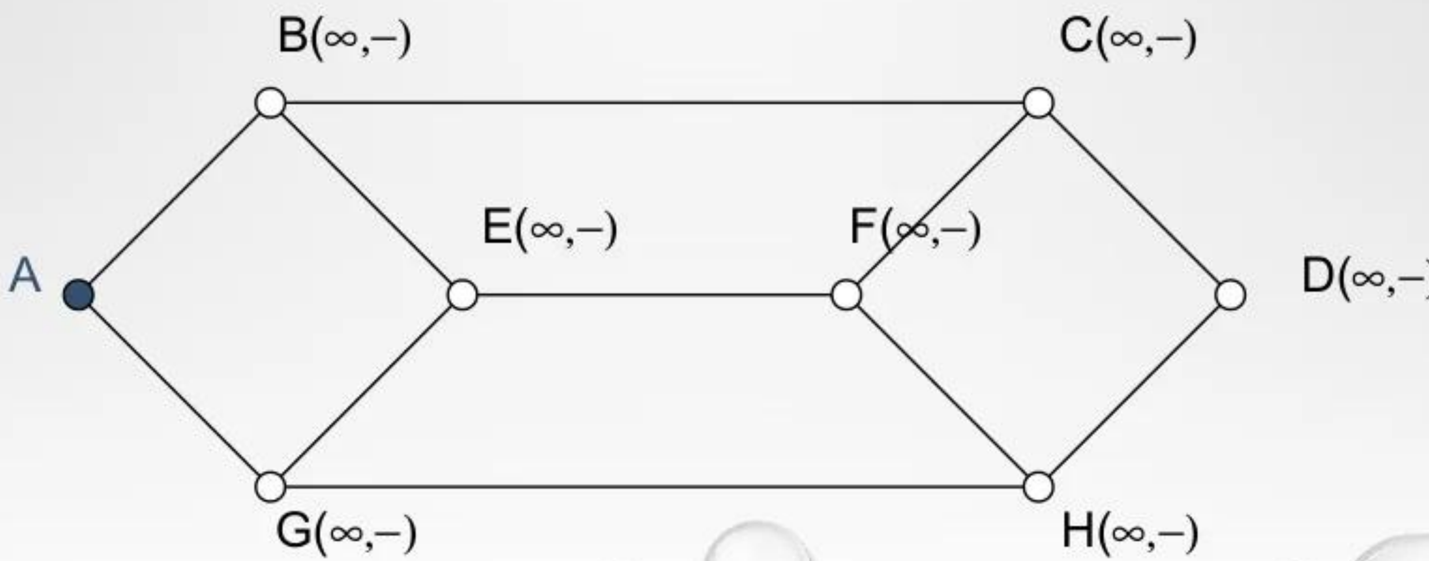
- We want to find the shortest path from A to D.
- Initially, no paths are known, so all nodes are labeled with infinity.



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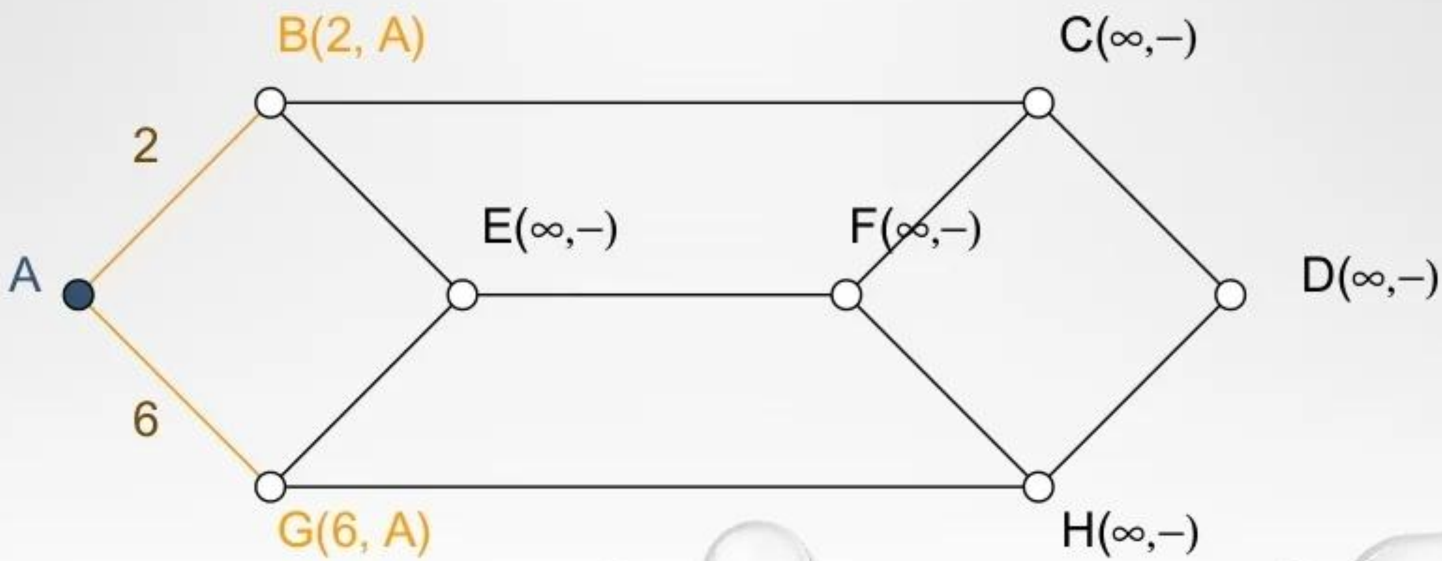
- We start out by marking node A (the working node) as permanent.
- We examine each of the nodes adjacent a, relabeling each one with the distance to a.



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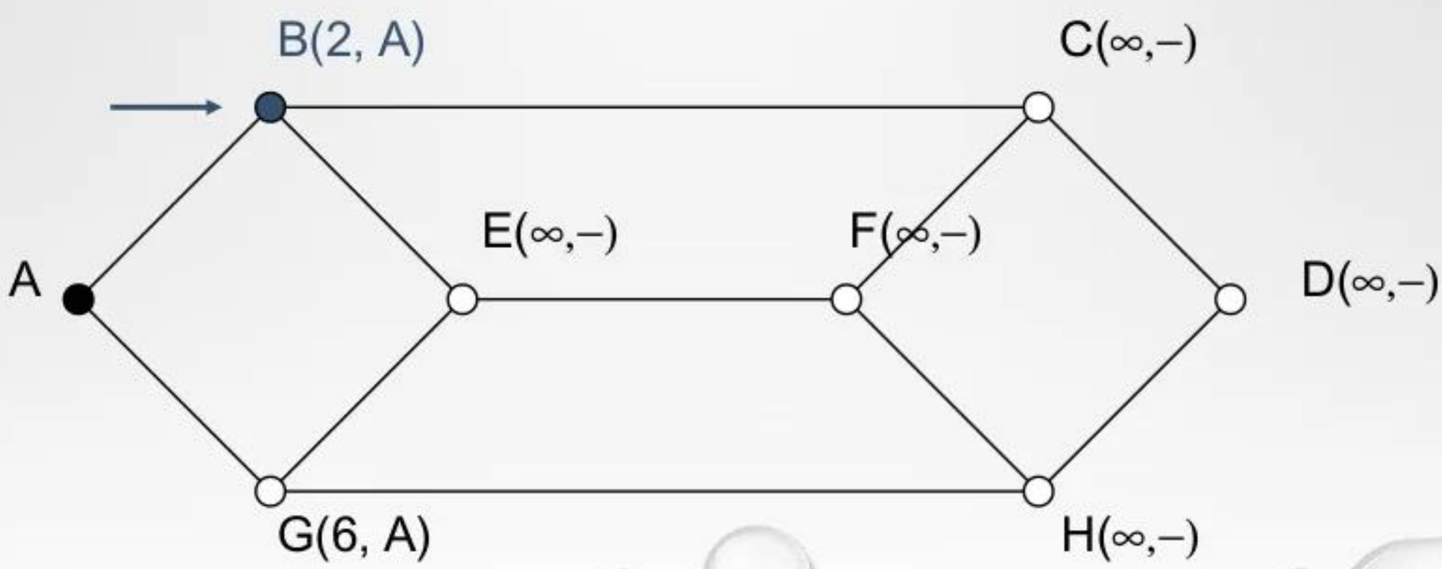
We make B with the smallest label permanent.

B becomes the new working node.



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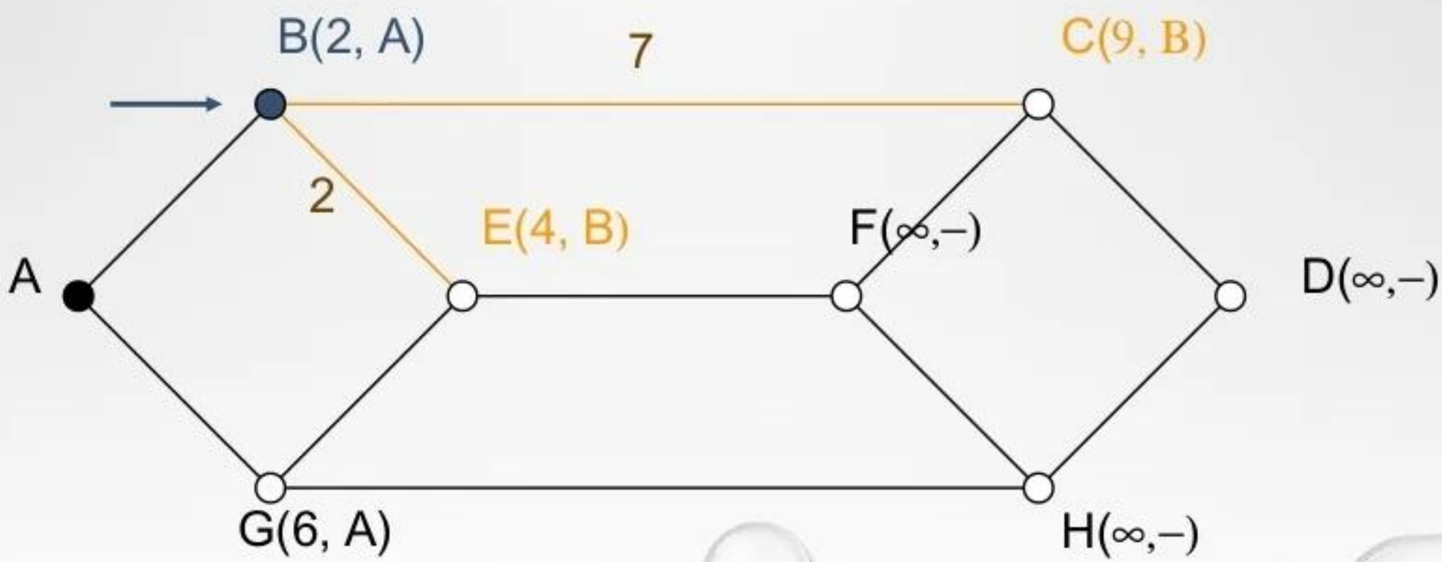
We examine each of the nodes adjacent B, relabeling each one with the distance to B.



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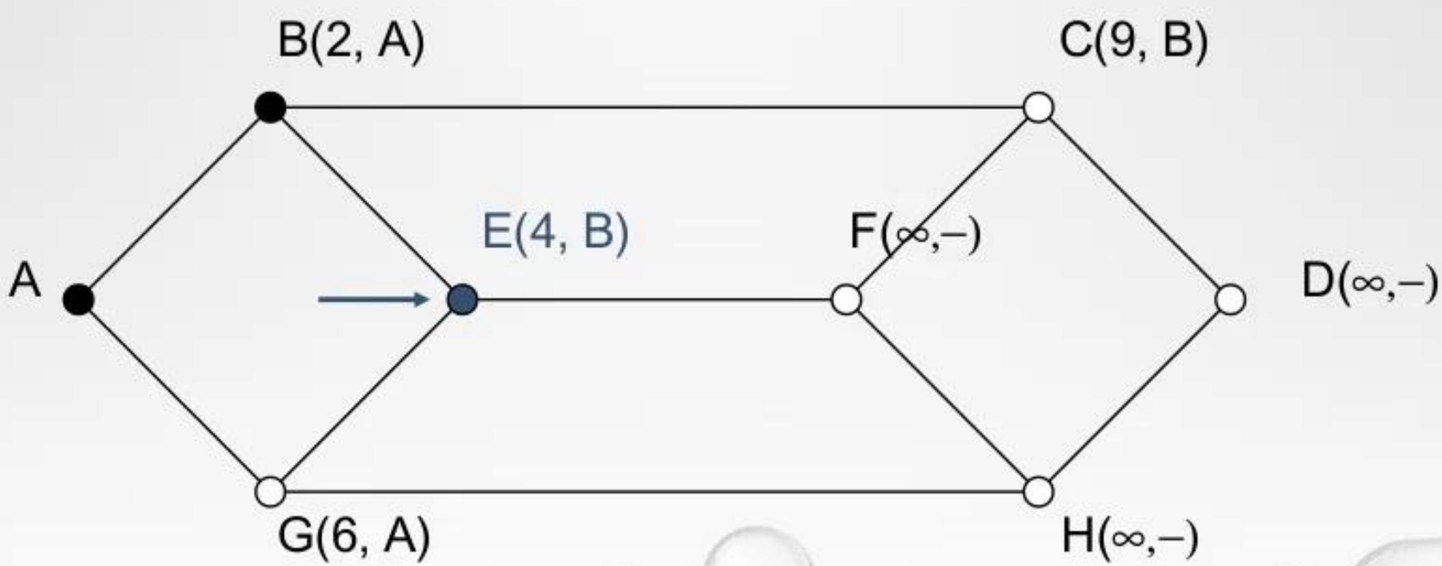
We make E with the smallest label permanent.

E becomes the new working node.



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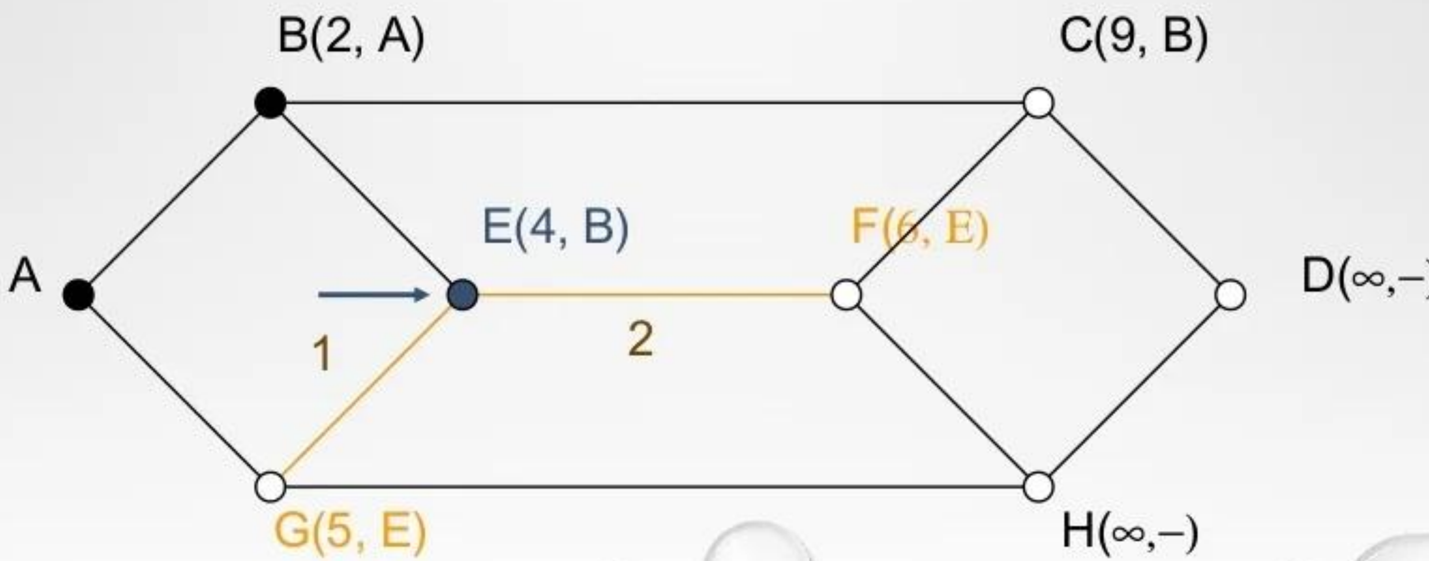
We examine each of the nodes adjacent E, relabeling each one with the distance to E.



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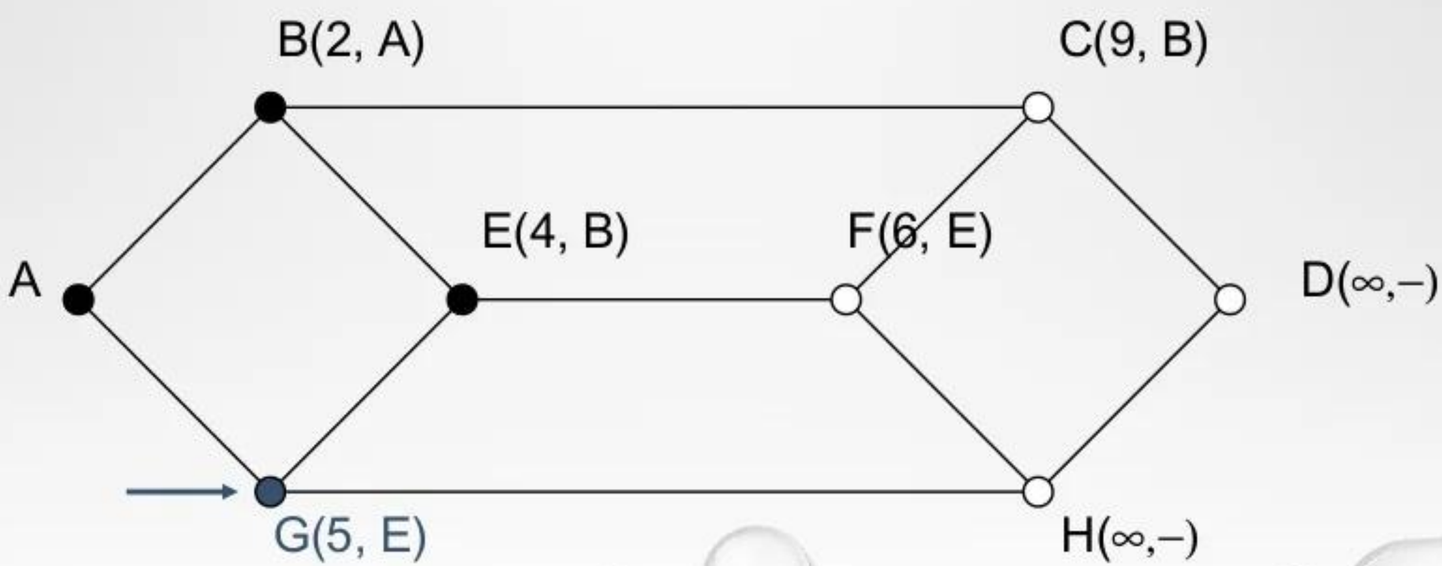
We make G with the smallest label permanent.

G becomes the new working node.



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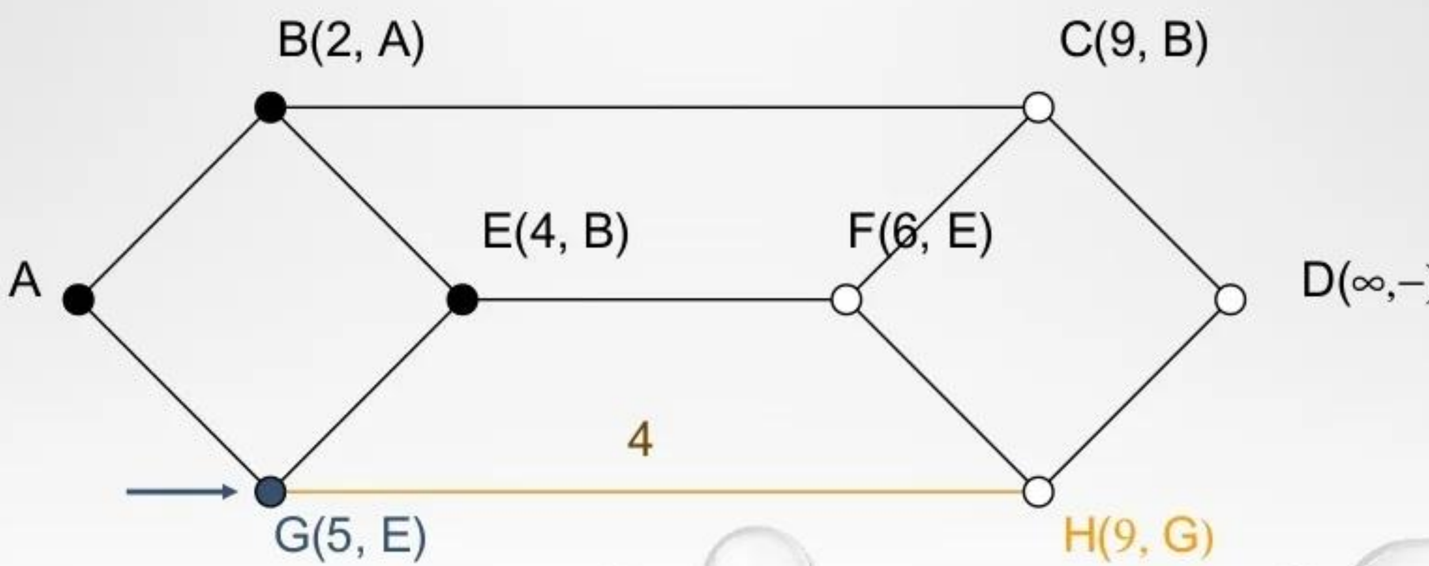
We examine each of the nodes adjacent G, relabeling each one with the distance to G.



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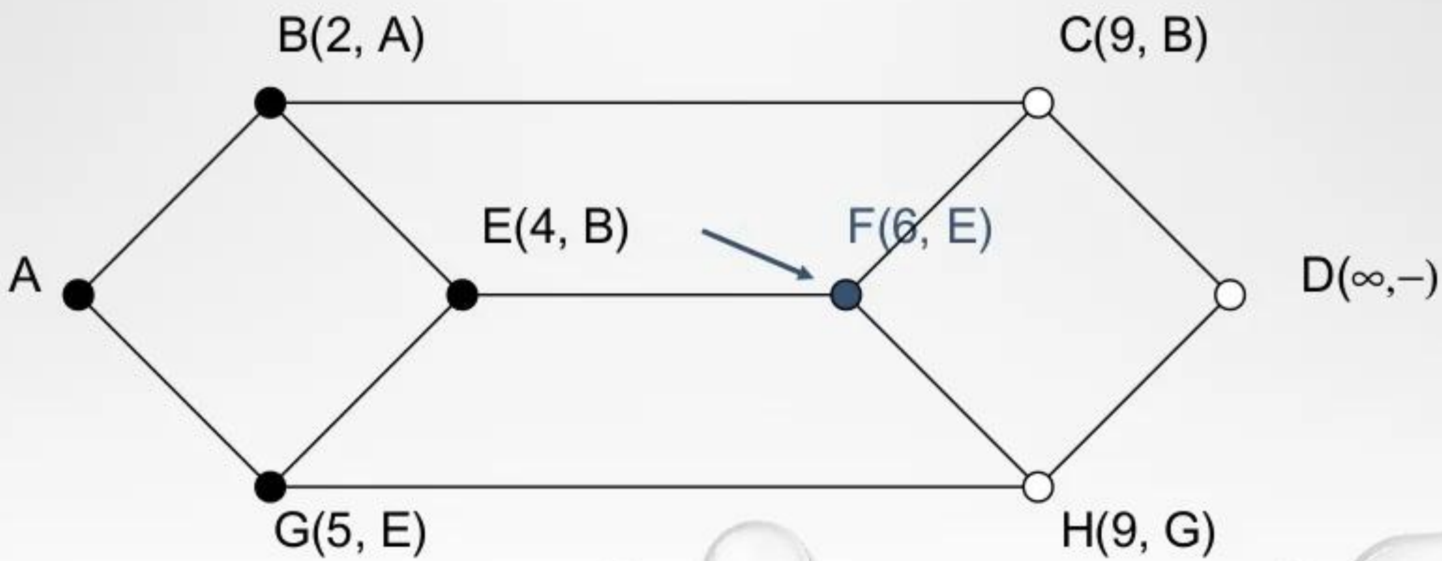
We make F with the smallest label permanent.

F becomes the new working node.



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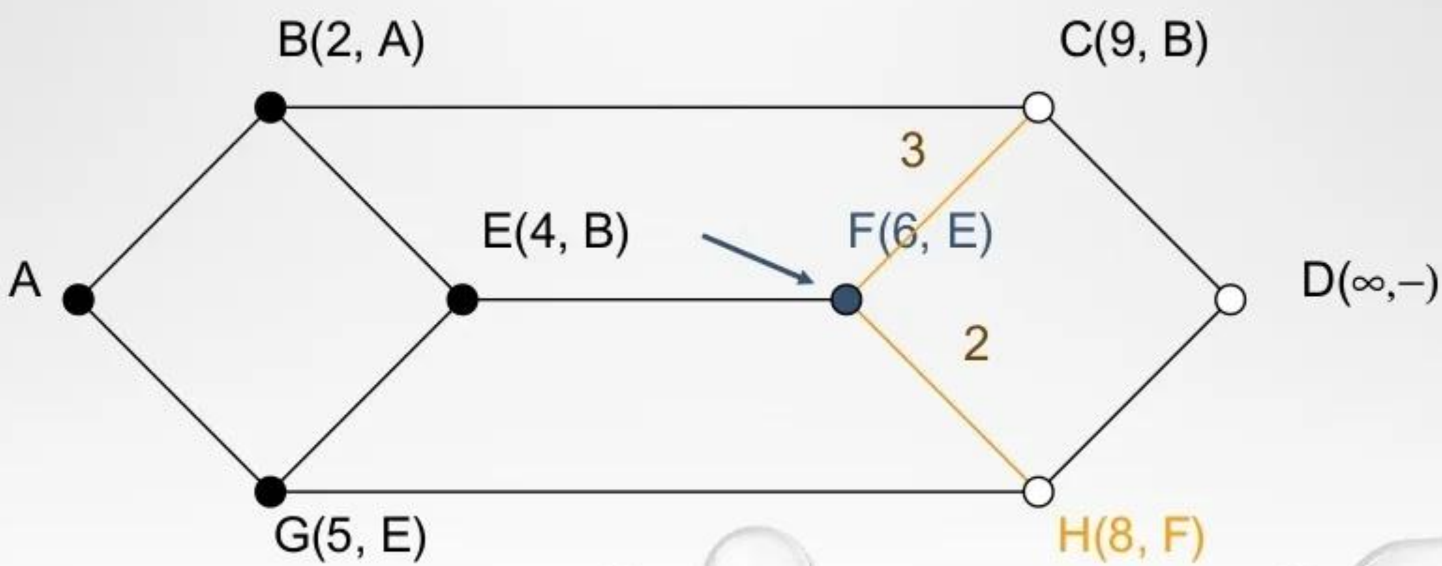
We examine each of the nodes adjacent F, relabeling each one with the distance to F.



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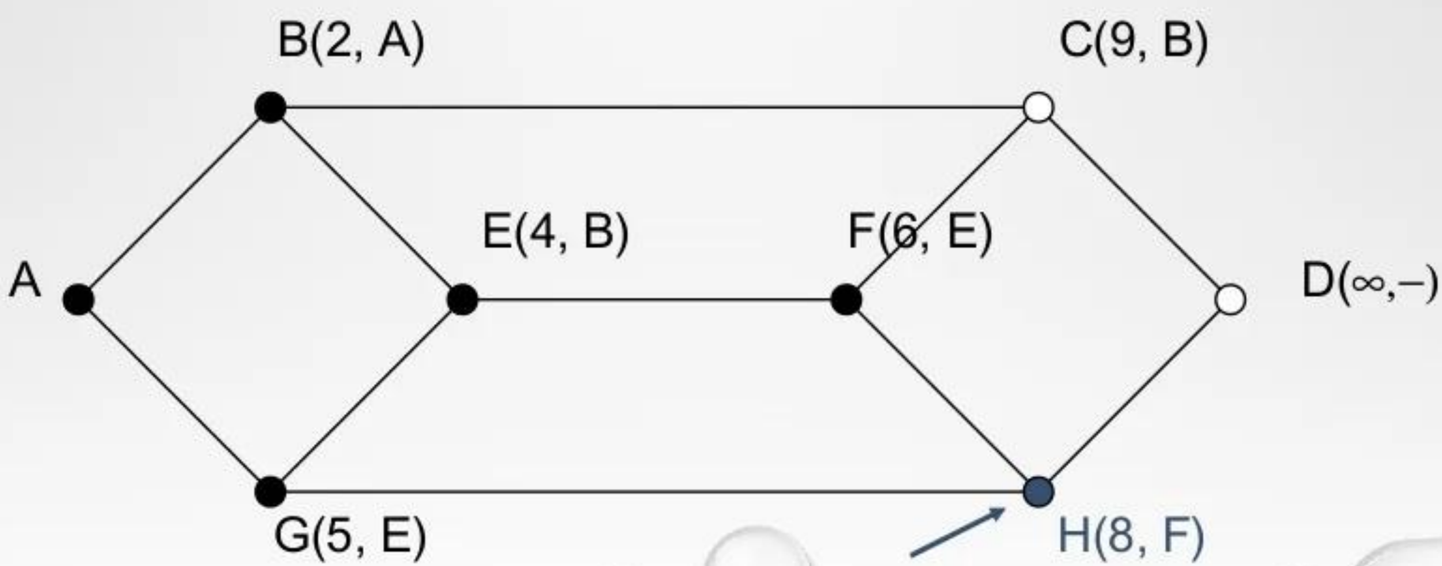
We make H with the smallest label permanent.

H becomes the new working node.



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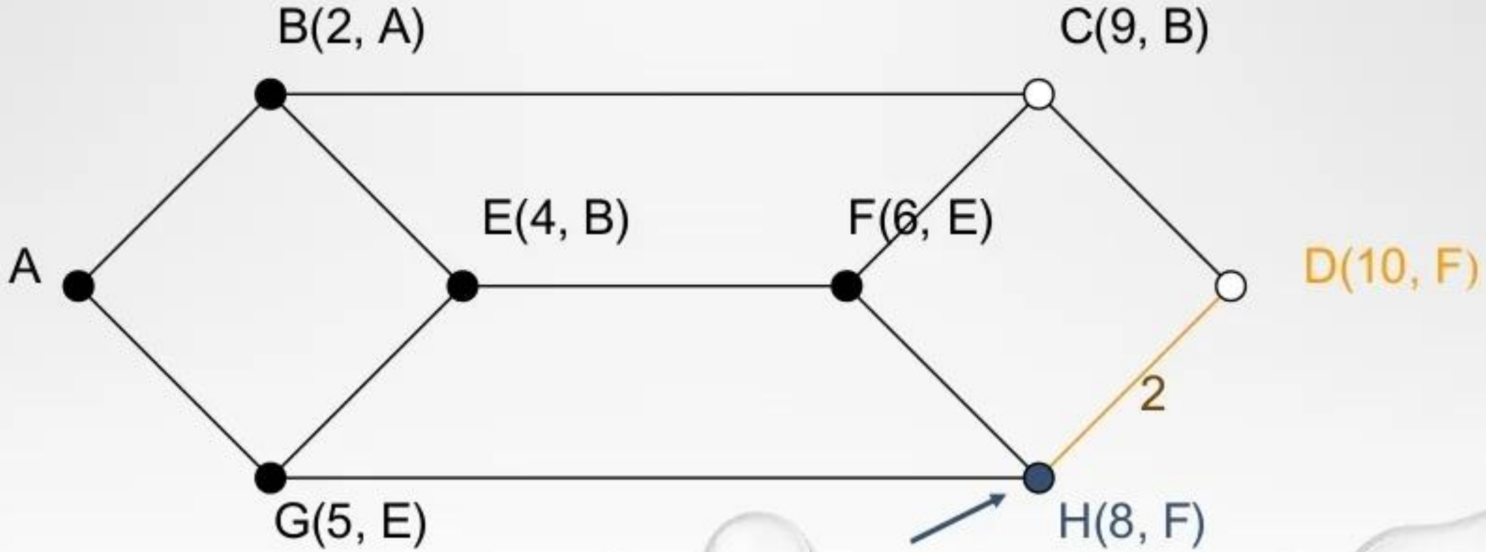
We examine each of the nodes adjacent H, relabeling each one with the distance to H.



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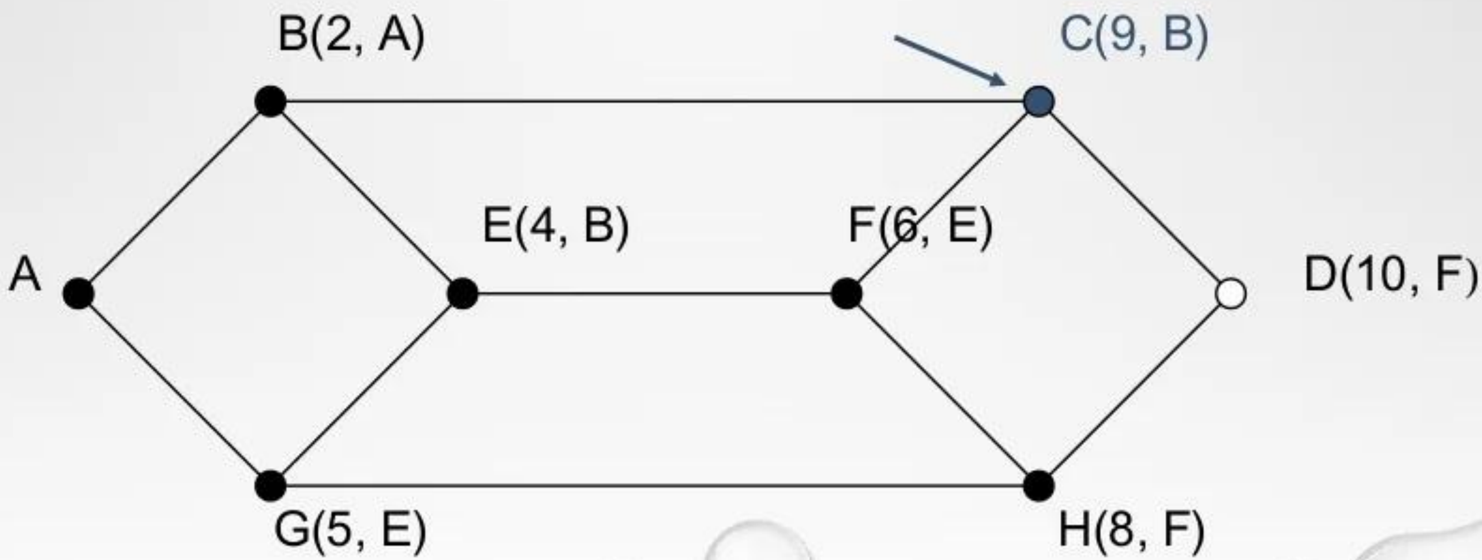
We make C with the smallest label permanent. C becomes the new working node.



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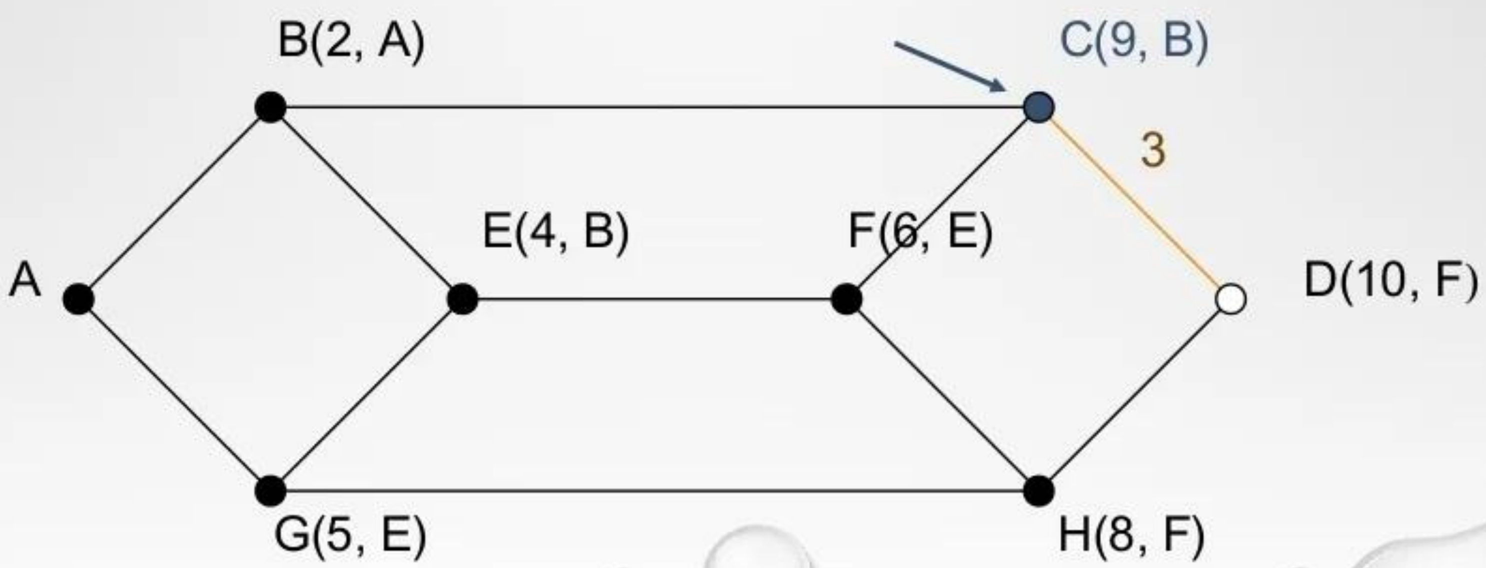
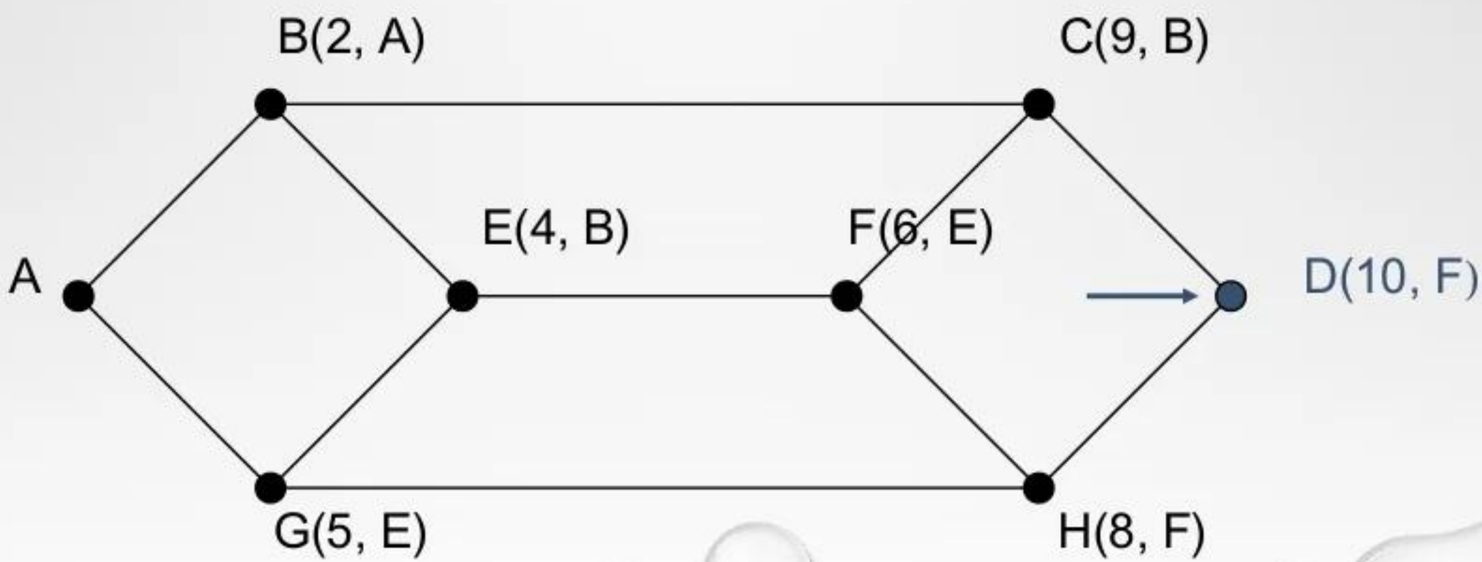
We examine each of the nodes adjacent C, relabeling each one with the distance to C.



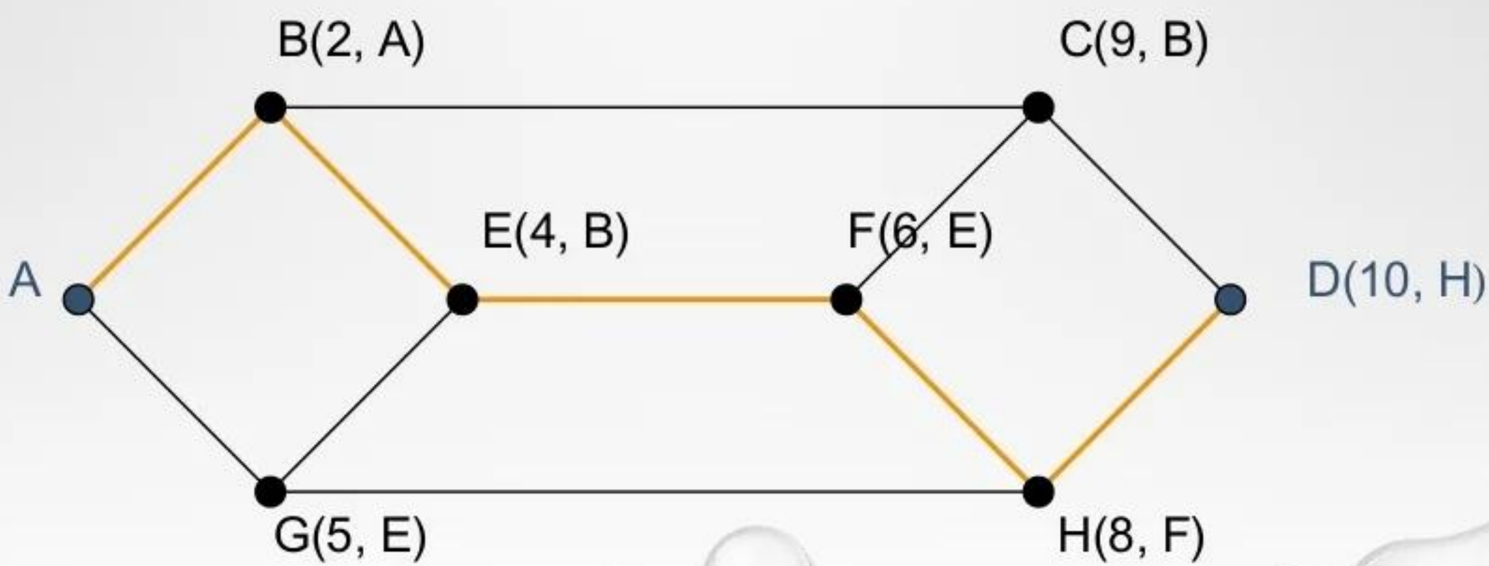
(Cont'd)

We make D with the smallest label permanent. D becomes the new working node.



(Cont'd)

The shortest path from A to D follows.



Distance Vector Routing	Link State Routing
Bellman ford algorithm is used in distance vector routing	Dijkstra's algorithm is used in link-state routing
It is simple to use	It needs trained network administrators
Chances of traffic are less	There is more chance of traffic
Convergence time is moderate i.e good news forward fastly as compared to bad news	Convergence time is fast
There is less utilisation of CPU and memory	There is more utilisation of CPU and memory
There is a problem of persistent looping	Only transient looping occurs
Best path is determined by the least number of hops	Best path is determined by the least cost
There is no hierarchical structure	There is a hierarchical structure
Require smaller bandwidth as there is no flooding, small packet, and local sharing	Require larger bandwidth for flooding problems and for transmitting large link state packets
It updates tables with information about its neighbors. So, it works based on local information	It has the information of the whole network. So, it works based on global information
It updates on a broadcast basis	It updates on a multicast basis

Network Layer Protocols

Various Network Layer protocols are as follows:

- IP
- ARP
- RARP
- ICMP
- DHCP
- BOOTP

Internet Protocol (IP)

- IP stands for **internet protocol**. It is a protocol defined in the TCP/IP model used for sending the packets from source to destination. The main task of IP is to deliver the packets from source to the destination based on the IP addresses available in the packet headers. IP defines the packet structure that hides the data which is to be delivered as well as the addressing method that labels the datagram with a source and destination information.
- An IP protocol provides the connectionless service, which is accompanied by two transport protocols, i.e., TCP/IP and UDP/IP, so internet protocol is also known as TCP/IP or UDP/IP.
- The first version of IP (Internet Protocol) was IPv4. After IPv4, IPv6 came into the market, which has been increasingly used on the public internet since 2006.

Function

The main function of the internet protocol is to provide addressing to the hosts, encapsulating the data into a packet structure, and routing the data from source to the destination across one or more IP networks. In order to achieve these functionalities, internet protocol provides two major things which are given below.

An internet protocol defines two things:

- Format of IP packet
- IP Addressing system

Before an IP packet is sent over the network, two major components are added in an IP packet, i.e., **header** and a **payload**.

- An IP header contains lots of information about the IP packet which includes:
- Source IP address: The source is the one who is sending the data.
- Destination IP address: The destination is a host that receives the data from the sender.
- Header length
- Packet length
- TTL (Time to Live): The number of hops occurs before the packet gets discarded.
- Transport protocol: The transport protocol used by the internet protocol, either it can be TCP or UDP.
- There is a total of 14 fields exist in the IP header, and one of them is optional.
- **Payload:** Payload is the data that is to be transported.

IP routing

IP routing is a process of determining the path for data so that it can travel from the source to the destination. As we know that the data is divided into multiple packets, and each packet will pass through a web of the router until it reaches the final destination. The path that the data packet follows is determined by the routing algorithm. The routing algorithm considers various factors like the size of the packet and its header to determine the efficient route for the data from the source to the destination. When the data packet reaches some router, then the source address and destination address are used with a routing table to determine the next hop's address. This process goes on until it reaches the destination. The data is divided into multiple packets so all the packets will travel individually to reach the destination.

IP Addressing

An IP address is a unique identifier assigned to the computer which is connected to the internet. Each IP address consists of a series of characters like 192.168.1.2. Users cannot access the domain name of each website with the help of these characters, so DNS resolvers are used that convert the human-readable domain names into a series of characters. Each IP packet contains two addresses, i.e., the IP address of the device, which is sending the packet, and the IP address of the device which is receiving the packet.

Types of IP addresses:

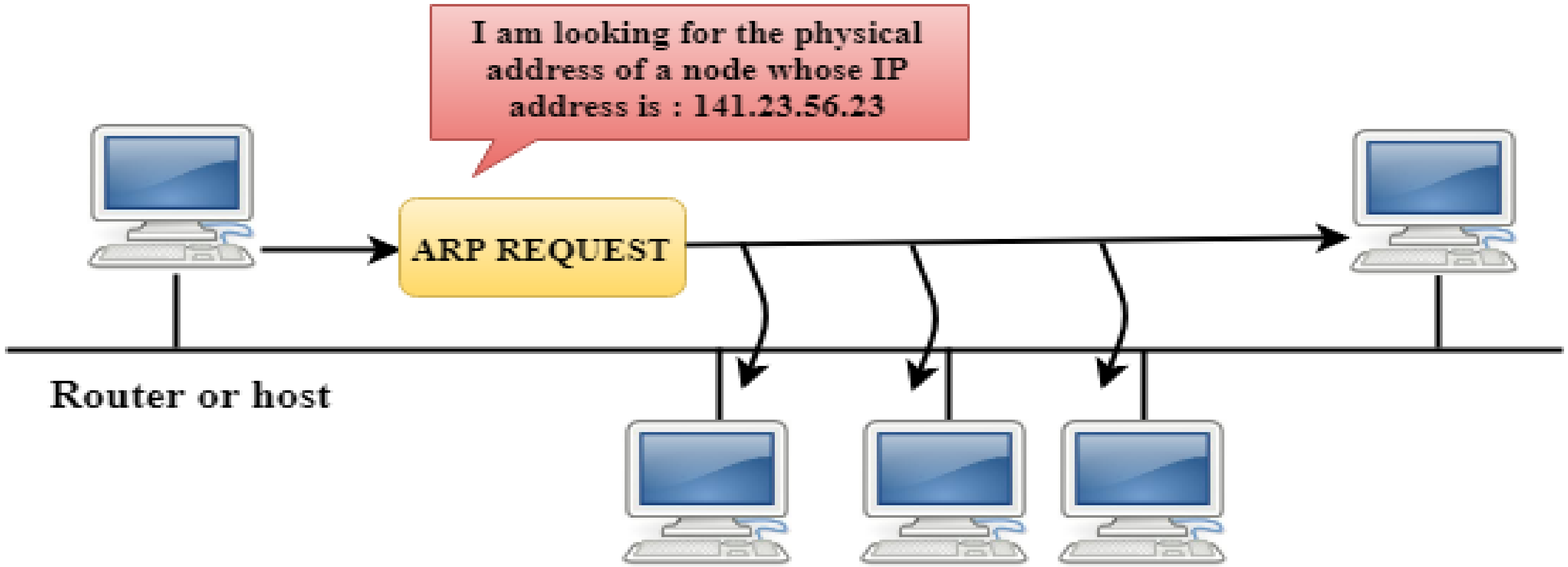
- **IPv4**
- **IPv6**

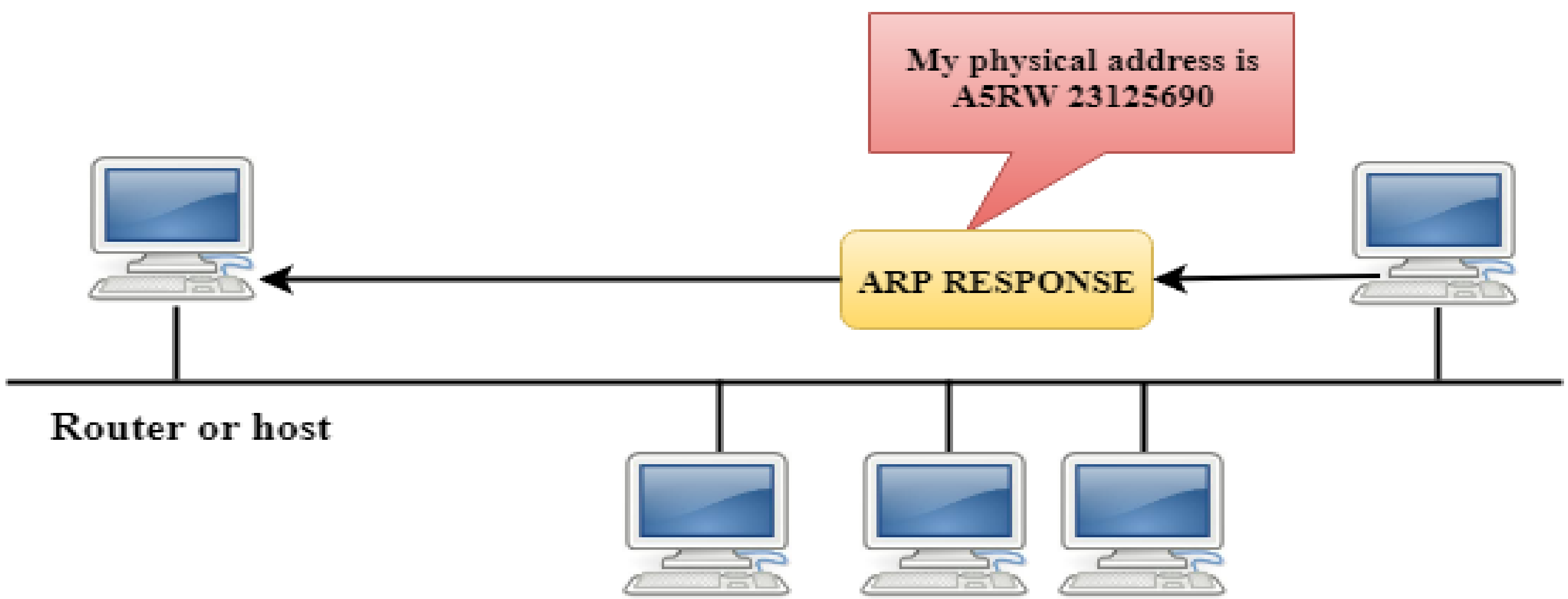
Address Resolution Protocol (ARP)

- ARP stands for Address Resolution Protocol.
- It is used to associate an IP address with the MAC address.
- Each device on the network is recognized by the MAC address imprinted on the NIC. Therefore, we can say that devices need the MAC address for communication on a local area network. MAC address can be changed easily. For example, if the NIC on a particular machine fails, the MAC address changes but IP address does not change. ARP is used to find the MAC address of the node when an internet address is known.

How ARP works

If the host wants to know the physical address of another host on its network, then it sends an ARP query packet that includes the IP address and broadcast it over the network. Every host on the network receives and processes the ARP packet, but only the intended recipient recognizes the IP address and sends back the physical address. The host holding the datagram adds the physical address to the cache memory and to the datagram header, then sends back to the sender.


I am looking for the physical address of a node whose IP address is : 141.23.56.23
ARP REQUEST
Router or host


My physical address is A5RW 23125690
ARP RESPONSE
Router or host

Steps taken by ARP protocol

If a device wants to communicate with another device, the following steps are taken by the device:

- The device will first look at its internet list, called the ARP cache to check whether an IP address contains a matching MAC address or not. It will check the ARP cache in command prompt by using a command **arp-a**.
- If ARP cache is empty, then device broadcast the message to the entire network asking each device for a matching MAC address.
- The device that has the matching IP address will then respond back to the sender with its MAC address
- Once the MAC address is received by the device, then the communication can take place between two devices.
- If the device receives the MAC address, then the MAC address gets stored in the ARP cache. We can check the ARP cache in command prompt by using a command arp -a.

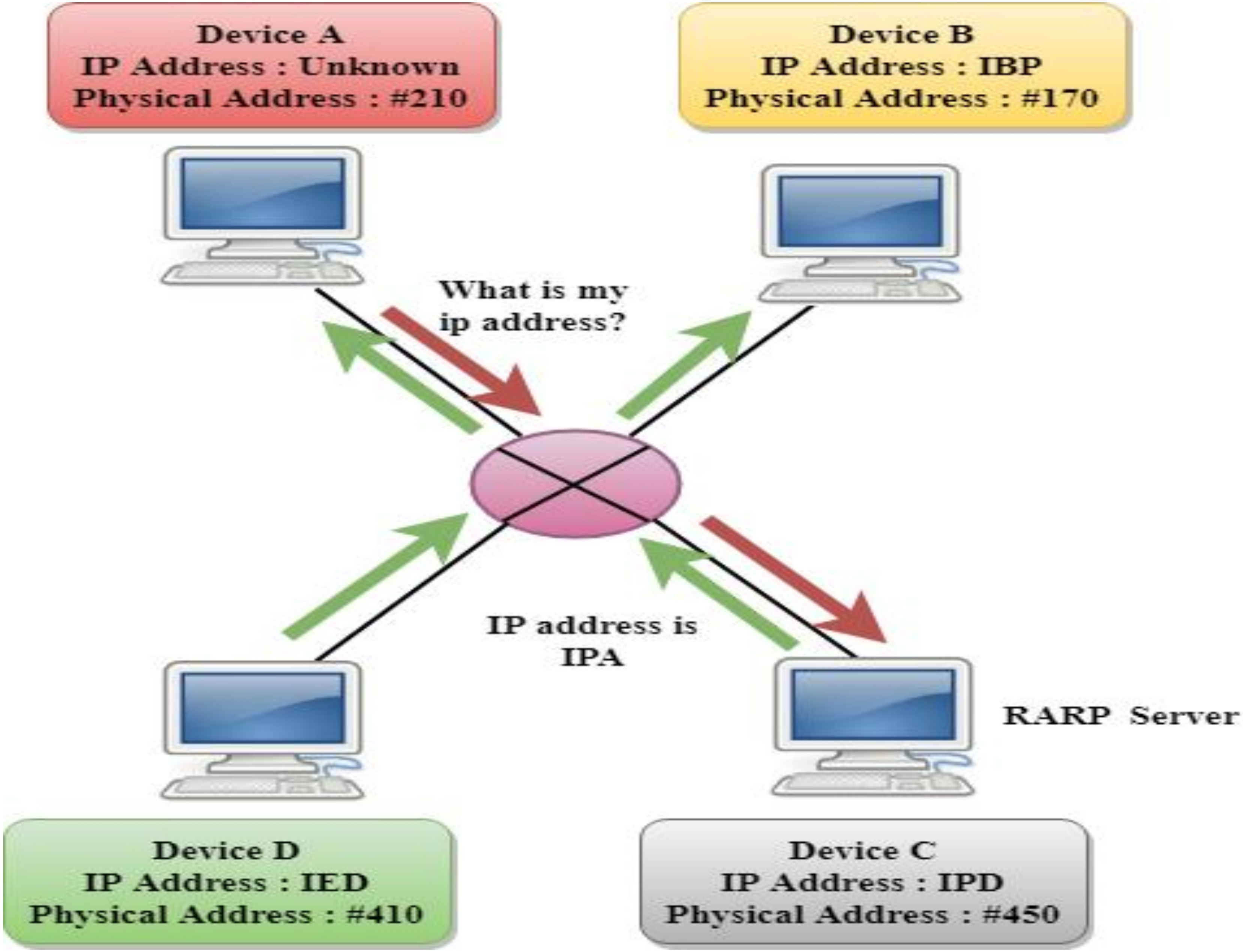
There are two types of ARP entries:

- **Dynamic entry:** It is an entry which is created automatically when the sender broadcast its message to the entire network. Dynamic entries are not permanent, and they are removed periodically.

- **Static entry:** It is an entry where someone manually enters the IP to MAC address association by using the ARP command utility.

Reverse Address Resolution Protocol (RARP)

- If the host wants to know its IP address, then it broadcast the RARP query packet that contains its physical address to the entire network. A RARP server on the network recognizes the RARP packet and responds back with the host IP address.
- The protocol which is used to obtain the IP address from a server is known as **Reverse Address Resolution Protocol**.
- The message format of the RARP protocol is similar to the ARP protocol.
- Like ARP frame, RARP frame is sent from one machine to another encapsulated in the data portion of a frame.

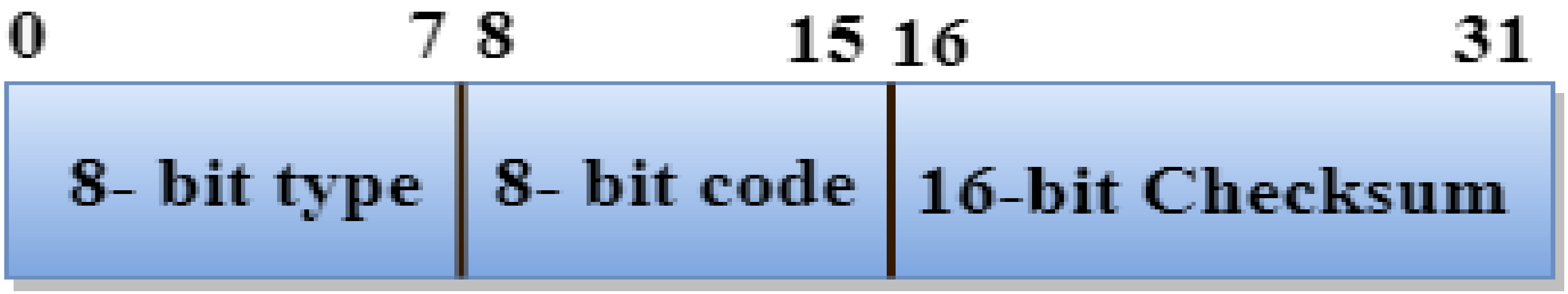

Device A
IP Address : Unknown
Physical Address : #210
Device B
IP Address : IBP
Physical Address : #170
What is my
ip address?
IP address is
IPA
RARP Server
Device D
IP Address : IED
Physical Address : #410
Device C
IP Address : IPD
Physical Address : #450

Internet Control Message Protocol (ICMP)

- ICMP stands for Internet Control Message Protocol.
- The ICMP is a network layer protocol used by hosts and routers to send the notifications of IP datagram problems back to the sender.
- ICMP uses echo test/reply to check whether the destination is reachable and responding.
- ICMP handles both control and error messages, but its main function is to report the error but not to correct them.

- An IP datagram contains the addresses of both source and destination, but it does not know the address of the previous router through which it has been passed. Due to this reason, ICMP can only send the messages to the source, but not to the immediate routers.
- ICMP protocol communicates the error messages to the sender. ICMP messages cause the errors to be returned back to the user processes.
- ICMP messages are transmitted within IP datagram.

The Format of an ICMP message



- The first field specifies the type of the message.
- The second field specifies the reason for a particular message type.
- The checksum field covers the entire ICMP message.

Five types of errors are handled by the ICMP protocol:

- **Destination unreachable:** The message of "Destination Unreachable" is sent from receiver to the sender when destination cannot be reached, or packet is discarded when the destination is not reachable.
- **Source Quench:** The purpose of the source quench message is congestion control. The message sent from the congested router to the source host to reduce the transmission rate. ICMP will take the IP of the discarded packet and then add the source quench message to the IP datagram to inform the source host to reduce its transmission rate. The source host will reduce the transmission rate so that the router will be free from congestion.
- **Time Exceeded:** Time Exceeded is also known as "Time-To-Live(TTL)". It is a parameter that defines how long a packet should live before it would be discarded.

There are two ways when Time Exceeded message can be generated:

- Sometimes packet discarded due to some bad routing implementation, and this causes the looping issue and network congestion. Due to the looping issue, the value of TTL keeps on decrementing, and when it reaches zero, the router discards the datagram. However, when the datagram is discarded by the router, the time exceeded message will be sent by the router to the source host.
- When destination host does not receive all the fragments in a certain time limit, then the received fragments are also discarded, and the destination host sends time Exceeded message to the source host.

- **Parameter problems:** When a router or host discovers any missing value in the IP datagram, the router discards the datagram, and the "parameter problem" message is sent back to the source host.

- **Redirection:** Redirection message is generated when host consists of a small routing table. When the host consists of a limited number of entries due to which it sends the datagram to a wrong router. The router that receives a datagram will forward a datagram to a correct router and also sends the "Redirection message" to the host to update its routing table.

Dynamic Host Configuration Protocol (DHCP)

- The Dynamic Host Configuration Protocol (DHCP) is a network management protocol used on Internet Protocol (IP) networks, whereby a DHCP server dynamically assigns an IP address and other network configuration parameters to each device on the network, so they can communicate with other IP networks. A DHCP server enables computers to request IP addresses and networking parameters automatically from the Internet service provider (ISP), reducing the need for a network administrator or a user to manually assign IP addresses to all network devices. In the absence of a DHCP server, a computer or other device on the network needs to be manually assigned an IP address.

- DHCP can be implemented on networks ranging in size from home networks to large campus networks and regional ISP networks. A router or a residential gateway can be enabled to act as a DHCP server. Most residential network routers receive a globally unique IP address within the ISP network. Within a local network, a DHCP server assigns a local IP address to each device connected to the network.

Allocating IP Addresses

Depending on implementation, the DHCP server may have three methods of allocating IP addresses:

- **Dynamic allocation**: A network administrator reserves a range of IP addresses for DHCP, and each DHCP client on the LAN is configured to request an IP address from the DHCP server during network initialization. The request-and-grant process uses a lease concept with a controllable time period, allowing the DHCP server to reclaim and then reallocate IP addresses that are not renewed.
- **Automatic allocation**: The DHCP server permanently assigns an IP address to a requesting client from the range defined by the administrator. This is like dynamic allocation, but the DHCP server keeps a table of past IP address assignments, so that it can preferentially assign to a client the same IP address that the client previously had.

- **Manual allocation**: Also commonly called *static allocation* and *reservations*. The DHCP server issues a private IP address dependent upon each client's *client id* based on a predefined mapping by the administrator.

- DHCP is used for Internet Protocol version 4 (IPv4) and IPv6.

Bootstrap Protocol (BOOTP)

- The **Bootstrap Protocol** (**BOOTP**) is a computer networking protocol used in Internet Protocol networks to automatically assign an IP address to network devices from a configuration server.
- When a computer that is connected to a network is powered up and boots its operating system, the system software broadcasts BOOTP messages onto the network to request an IP address assignment. A BOOTP configuration server assigns an IP address based on the request from a pool of addresses configured by an administrator. BOOTP operates only on IPv4 networks.

Operation

Case 1 : Client and server on same network

When a BOOTP client is started, it has no IP address, so it broadcasts a message containing its MAC address onto the network. This message is called a "BOOTP request," and it is picked up by the BOOTP server, which replies to the client with the following information that the client needs:

1. The client's IP address, subnet mask, and default gateway address
2. The IP address and host name of the BOOTP server
3. The IP address of the server that has the boot image, which the client needs to load its operating system

Case 2 : Client and server on different networks

- Problem with the bootp request is that the request is broadcast. A broadcast IP data-gram cannot pass through any router. The router discards this packet.
- To solve this problem there is a need for an intermediary (relay).
- One of the host or router can be configured at application layer to operate as relay agent.
- The relay agent knows the uni-cast address of bootp server and listens for broadcast message on port 67.
- When it receives this broadcast packet, it encapsulates the message in uni-cast data-gram and sends request to bootp server.
- The packet carrying a uni-cast destination address is routed by any router and reaches the bootp server.
- The relay agent after receiving the reply, sends it to bootp client.

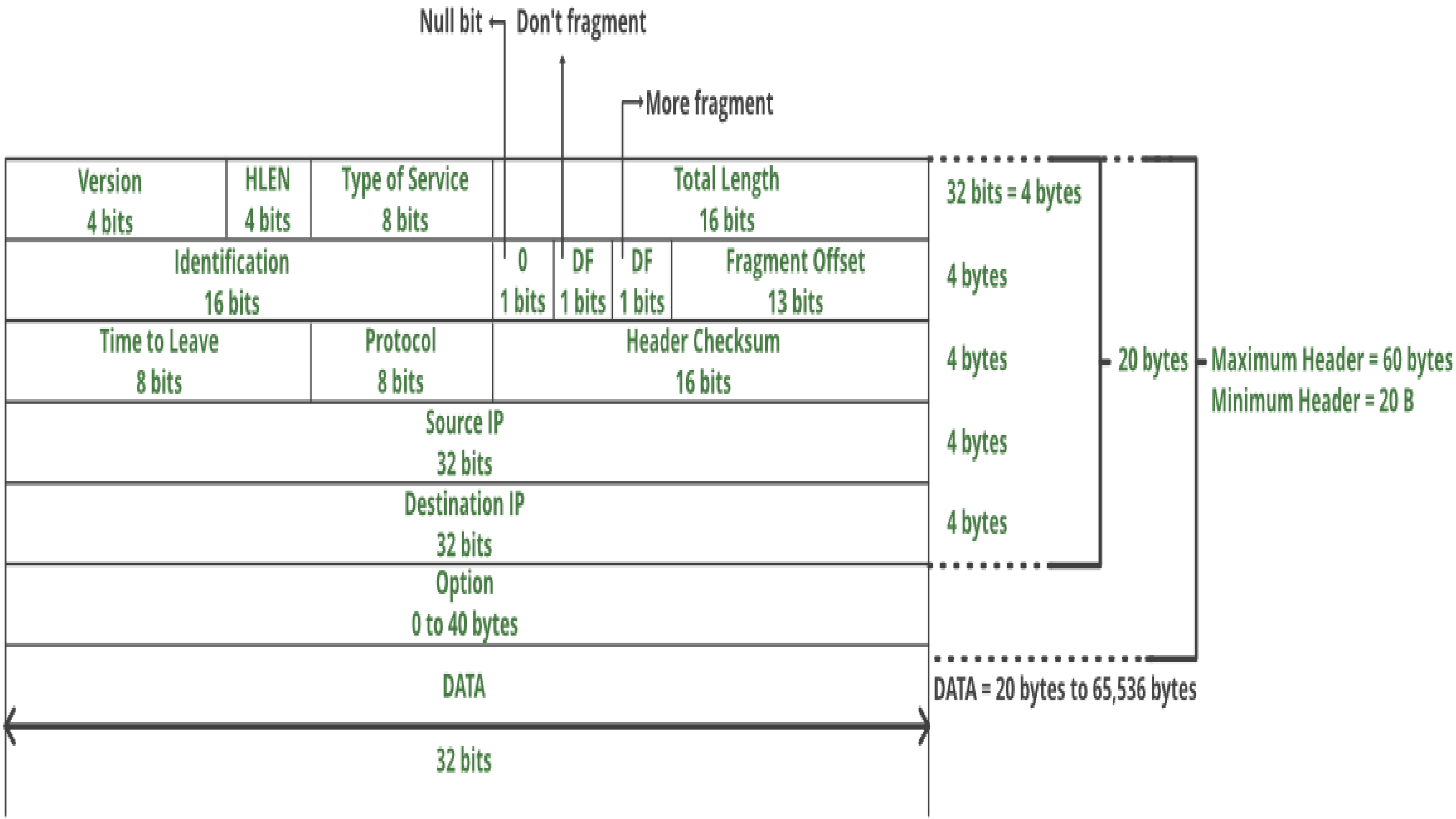
IPv4

IPv4 is a connectionless protocol used for packet-switched networks. It operates on a best effort delivery model, in which neither delivery is guaranteed, nor proper sequencing or avoidance of duplicate delivery is assured. Internet Protocol Version 4 (IPv4) is the fourth revision of the Internet Protocol and a widely used protocol in data communication over different kinds of networks. IPv4 is a connectionless protocol used in packet-switched layer networks, such as Ethernet. It provides a logical connection between network devices by providing identification for each device. There are many ways to configure IPv4 with all kinds of devices – including manual and automatic configurations – depending on the network type.

IPv4 uses 32-bit addresses for Ethernet communication in five classes: A, B, C, D and E. Classes A, B and C have a different bit length for addressing the network host. Class D addresses are reserved for military purposes, while class E addresses are reserved for future use. IPv4 uses 32-bit (4 byte) addressing, which gives 2^{32} addresses. IPv4 addresses are written in the dot-decimal notation, which comprises of four octets of the address expressed individually in decimal and separated by periods, for instance, 192.168.1.5.

IPv4 Datagram Header

Size of the header is 20 to 60 bytes.



- ***VERSION:*** *Version of the IP protocol (4 bits), which is 4 for IPv4*
- ***HLEN:*** *IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.*
- ***Type of service:*** *Low Delay, High Throughput, Reliability (8 bits)*
- ***Total Length:*** *Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.*
- ***Identification:*** *Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)*
- ***Flags:*** *3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)*
- ***Fragment Offset:*** *Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.*
- ***Time to live:*** *Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.*
- ***Protocol:*** *Name of the protocol to which the data is to be passed (8 bits)*
- ***Header Checksum:*** *16 bits header checksum for checking errors in the datagram header*
- ***Source IP address:*** *32 bits IP address of the sender*
- ***Destination IP address:*** *32 bits IP address of the receiver*
- ***Option:*** *Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.*

IPv6

IPv6 was developed by Internet Engineering Task Force (IETF) to deal with the problem of IP v4 exhaustion. IP v6 is 128-bits address having an address space of 2^128, which is way bigger than IPv4. In IPv6 we use Colon-Hexa representation. There are 8 groups and each group represents 2 Bytes.

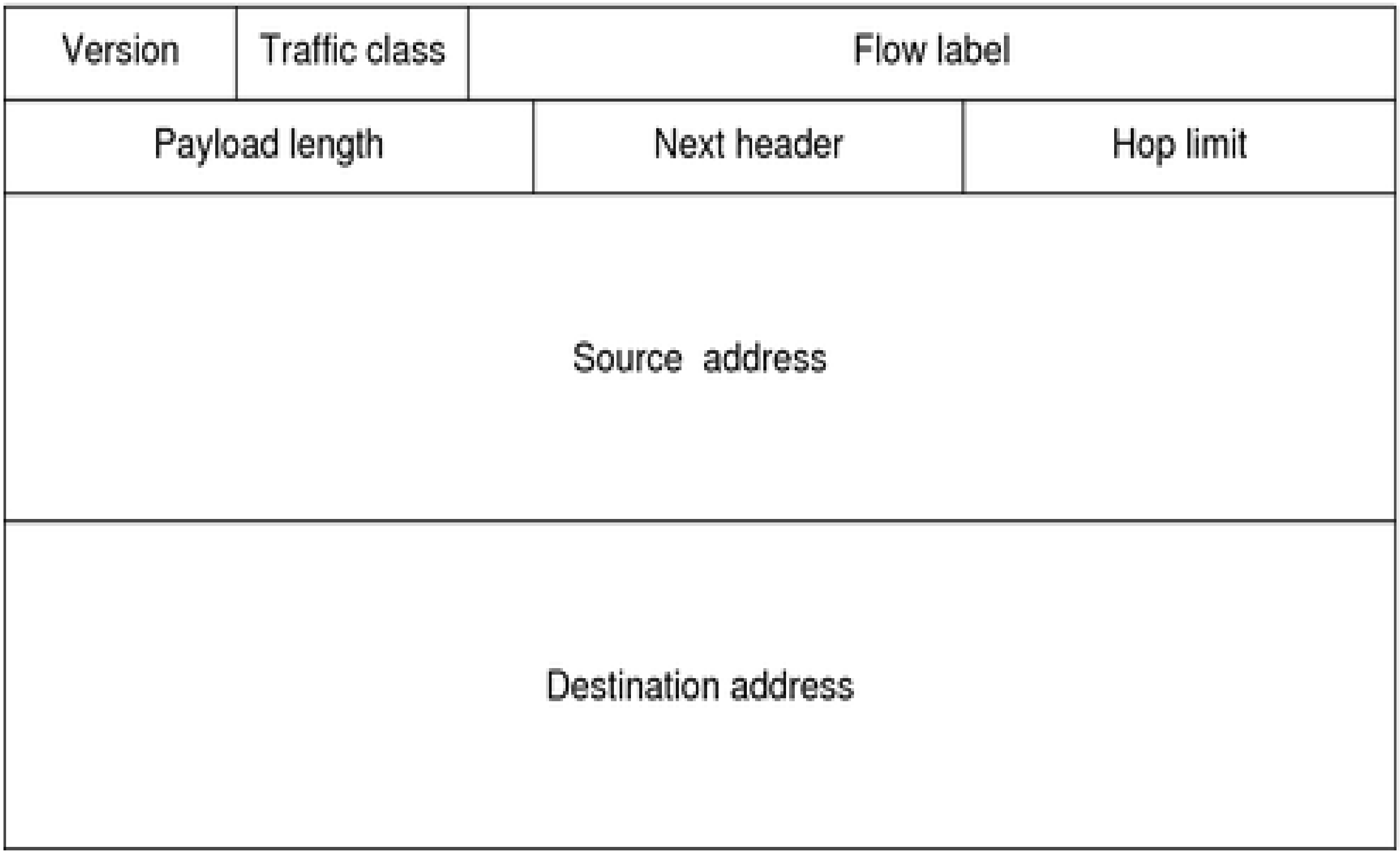
In IPv6 representation, we have three addressing methods :

- Unicast
- Multicast
- Anycast

- **Unicast Address:** Unicast Address identifies a single network interface. A packet sent to unicast address is delivered to the interface identified by that address.
- **Multicast Address:** Multicast Address is used by multiple hosts, called as Group, acquires a multicast destination address. These hosts need not be geographically together. If any packet is sent to this multicast address, it will be distributed to all interfaces corresponding to that multicast address.
- **Anycast Address:** Anycast Address is assigned to a group of interfaces. Any packet sent to anycast address will be delivered to only one member interface (mostly nearest host possible).

Note : Broadcast is not defined in IPv6.

IPv6 Frame Structure



The following list describes the function of each header field.

- **Version** – 4-bit version number of Internet Protocol = 6.
- **Traffic class** – 8-bit traffic class field.
- **Flow label** – 20-bit field.
- **Payload length** – 16-bit unsigned integer, which is the rest of the packet that follows the IPv6 header, in octets.
- **Next header** – 8-bit selector. Identifies the type of header that immediately follows the IPv6 header. Uses the same values as the IPv4 protocol field.
- **Hop limit** – 8-bit unsigned integer. Decremented by one by each node that forwards the packet. The packet is discarded if the hop limit is decremented to zero.
- **Source address** – 128 bits. The address of the initial sender of the packet.
- **Destination address** – 128 bits. The address of the intended recipient of the packet. The intended recipient is not necessarily the recipient if an optional routing header is present.

Difference Between IPv4 and IPv6

IPv4	IPv6
IPv4 has 32-bit address length	IPv6 has 128-bit address length
It Supports Manual and DHCP address configuration	It supports Auto and renumbering address configuration
In IPv4 end to end connection integrity is Unachievable	In IPv6 end to end connection integrity is Achievable
It can generate 4.29×109 address space	Address space of IPv6 is quite large it can produce 3.4×1038 address space
Security feature is dependent on application	IPSEC is inbuilt security feature in the IPv6 protocol
Address representation of IPv4 in decimal	Address Representation of IPv6 is in hexadecimal
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation performed only by sender

IPv4	IPv6
In IPv4 Packet flow identification is not available	In IPv6 packet flow identification are Available and uses flow label field in the header
In IPv4 checksum field is available	In IPv6 checksum field is not available
It has broadcast Message Transmission Scheme	In IPv6 multicast and any cast message transmission scheme is available
In IPv4 Encryption and Authentication facility not provided	In IPv6 Encryption and Authentication are provided
IPv4 has header of 20-60 bytes.	IPv6 has header of 40 bytes fixed

Logical Address

- Logical address are necessary for universal communications to identify each host in the internet for delivery of a packet from host to host.
- All IPv4 addresses are 32 bits long (equivalently 4 bytes).
- Within an IP address it encodes its network number and host number.
- The physical (network) addresses will change from hop to hop, but the logical addresses remains the same.

IPV4 Address

- IPv4 addresses are actually **32-bit** binary numbers, consisting of the two sub addresses (identifiers) mentioned above which, respectively, **identify the network and the host to the network**, with an imaginary boundary separating the two.
- An IP address is, as such, generally shown as **4 octets** of numbers from 0-255 represented in decimal form instead of binary form.
- An IPv4 address is typically expressed in **dotted-decimal notation**, with every eight bits (octet) represented by a number from **one to 255**, each separated by a dot.
- An example IPv4 address would look like this:

192.168.17.12

Network id and Host id

- Network ID is the portion of an IP address that identifies the TCP/IP network on which a host resides.
- The network ID portion of an IP address uniquely identifies the host's network on an internetwork, while the host ID portion of the IP address identifies the host within its network.
- Together, the host ID and network ID, which make up the entire IP address of a host, uniquely identify the host on a TCP/IP internetwork.

Address Space

- An address space is the total number of addresses used by the protocol.
- For an N bits address, 2^N bits address space can be used, because each bit can have two different values (0 or 1).

Notations

- Two types of notations in IPv4.

1. Binary Notation
2. Dotted- Decimal Notation

Binary Notation

- The binary notation in IPv4 is a 32 bit address or 4 bytes addres.

Ex;

01000000 00100000 00010000 00000111